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Establishing a Data Innovation Lab: A Practical Guide to Applying the AfDB's Data Innovation Maturity Model



Data Innovation for Africa

Data Innovation for Africa Initiative (DI4A)

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**Statistics Department
Economic Governance and Knowledge Management Complex
African Development Bank Group**

“Innovation does not mean having a new idea, but no longer having an old one”

Edwin H. Land



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Acronyms

AfDB	African Development Bank
AI	Artificial Intelligence
AIH	Africa Information Highway
ANSD	Agence Nationale de la Statistique et de la Démographie (Senegal)
ANSTAT	Agence Nationale de la Statistique (Côte d'Ivoire)
AU	African Union
CDO	Chief Data Officer
CI/CD	Continuous Integration / Continuous Delivery
COMESA	Common Market for Eastern and Southern Africa
DDP	Development Data Partnership
DI4A	Data Innovation for Africa Initiative
DIF	Data Innovation Fund
DIL	Data Innovation Lab
ECST	AfDB Statistics Department
ENSAE	École Nationale de la Statistique et de l'Analyse Économique (Senegal)
GDDS	General Data Dissemination System
GPU	Graphics Processing Unit
GSBPM	Generic Statistical Business Process Model
H100 / A100	NVIDIA GPU Models
HRSL	High Resolution Settlement Layer (Meta)
ICT	Information and Communication Technology
IaC	Infrastructure as Code
IMF	International Monetary Fund
INS	Institut National de la Statistique (Tunisia)
JBA	Jeremy Benn Associates Ltd.
LLM	Large Language Model
MLDA	Master Data License Agreement
MLOps	Machine Learning Operations
MoU	Memorandum of Understanding
NCC	National Communications Commission (Nigeria)
N-DEEDS	Nigeria Digital Economy Ecosystem Development Strategy
NDS	National Data Strategy
Ne-GIF	Nigeria e-Government Interoperability Framework
NNBP	Nigerian National Broadband Plan
NISR	National Institute of Statistics of Rwanda

NITDA	National Information Technology Development Agency (<i>implied via Nigeria references</i>)
NVIDIA	NVIDIA Corporation (<i>note: brand name, not acronym</i>)
NODP	Nigeria Open Data Policy
NSAI	National Strategy for Artificial Intelligence
NSDS	National Strategy for the Development of Statistics
NSO	National Statistical Office
NSS	National Statistical System
NTC	National Centre for Remote Sensing (Tunisia)
ODIN	Open Data Inventory
ODP	Open Data Platform
PARIS21	Partnership in Statistics for Development in the 21st Century
R&D	Research and Development
RECs	Regional Economic Communities
RMC	Regional Member Countries
SDD	Statistical Data Dissemination (<i>appears within context of SDDS and GDDS</i>)
SDDS	Special Data Dissemination Standard
SDG	Sustainable Development Goals
SDMX	Statistical Data and Metadata Exchange
STG17	Special Technical Group on Emerging Issues (AU STATAFRIC)
TFP	Technical and Financial Partners
UN	United Nations
UN-CEBD	UN Committee of Experts on Big Data and Data Science for Official Statistics
UNECA	United Nations Economic Commission for Africa
UNQAF	United Nations Quality Assurance Framework
UNSD	United Nations Statistics Department
VIIRS	Visible Infrared Imaging Radiometer Suite
WB	World Bank

Foreword

We are delighted to present this guide, which explores a topic of profound importance to the transformation of data ecosystems, particularly in contexts where National Statistical Offices (NSOs) serve as vital agents of change. In today's rapidly evolving information landscape, data innovation is no longer a choice; it is a strategic necessity. This guide serves as a practical resource for those navigating the complex challenges facing official statistics and aims to leverage innovation as a transformative force to enhance the availability, quality, and relevance of data.

This guide is built on a simple but powerful conviction: data innovation is not a luxury, it is a strategic investment. By embracing it, NSOs and their partners can stay agile, adapt quickly, and thrive in the face of rapid technological advancements and digital transformation.

This guide draws on early insights from the **Data Innovation Lab (DIL)** initiative, launched in 2023 by the **African Development Bank (AfDB)** as a core component of its **Africa Information Highway (AIH)** program. As the demand for high-quality development data in Africa continues to expand – and as patterns of data use evolve alongside rapid technological advances – statistical institutions are being called upon to fundamentally rethink their operational models, practices as well as the legal and strategic frameworks that have traditionally governed the production and use of statistics.

To support the process of remodeling statistical production activities in the context of a rapidly evolving digital landscape, the guide addresses four key questions:

- What does innovation in statistics entail?
- How can innovation in statistics enhance our ability to meet the evolving needs of data users?
- What dimensions should be considered in developing an effective data innovation action plan?
- How can we assess the readiness of data ecosystems to undergo effective transformation?

Designed to be both actionable and forward-looking, the guide features practical examples from countries that have participated in the DIL initiative. As the initiative progresses, we anticipate that this guide will continue to expand – enriched by diverse experiences, new insights, and shared learning across the continent.

The **Statistics Department of the AfDB** is proud to lead the DIL initiative, and to support African countries in advancing their innovative, bold, and forward-looking visions for statistical transformation. We hope this guide serves as a source of inspiration and practical guidance, encouraging new ways of thinking, reducing uncertainties, and empowering statisticians across Africa to thrive in an increasingly dynamic and demanding data ecosystem.

We wish you an engaging and insightful read!

Babatunde Samson OMOTOSHO
Director, Statistics Department

About this guide

This guide is the product of ongoing reflections within the Data Innovation Team of the AfDB's Statistics Department (ECST), following the launch of the DIL initiative in 2023. The guide is designed to serve two main purposes.

- First, to present the DIL initiative and its results. The guide provides a comprehensive overview of the DIL initiative – its conceptual foundations, milestones, and early achievements. It highlights the catalytic interventions that have supported the establishment of Data Innovation Labs in several African countries. These efforts have fostered valuable insights and experiences, which this guide seeks to share with National Statistical Offices (NSOs) and their partners as they navigate the evolving landscape of data innovation across Africa.
- Second, to provide guidance to the NSOs and partners on how to embed innovation. This guide responds to the growing interest among African NSOs to integrate innovation into their institutional processes. It explores how such integration can transform national data ecosystems, enhance responsiveness, and drive greater impact. With many NSOs already committed - or planning - to establish Data Innovation Labs, the guide offers practical guidance, organized around frequently asked questions and real-world challenges, and informed by existing experiences.

The AfDB continues to play a pivotal role in advancing statistical innovation across the continent. Since the late 2023, the Bank has conducted diagnostic missions in Tunisia (November 2023), Senegal (February 2024), Côte d'Ivoire (March 2024), Nigeria (June 2024), Cameroon (September 2024), and Zambia (April 2025). These missions assessed each country's readiness for implementing data innovation and laid the groundwork for tailored action plans. Progress in these projects is continuously monitored, allowing for the identification of best practices and the promotion of peer learning across countries.

The AfDB's Statistics Department extends its deep appreciation to the countries that have piloted its DIL maturity model and embarked on the journey of transforming their data ecosystems. This is a long-term endeavor requiring sustained commitment, and the AfDB remains dedicated to supporting these efforts—both through its own resources and by helping mobilize additional technical and financial support.

This guide was developed by the AfDB's Data Innovation Team, comprising consultants Thierry PACCOUD and Mohamed Ali HADJTAIEB, under the coordination of Momar KOUTA, Senior Statistical Information Systems Officer. It was produced under the supervision of Louis Koua KOUAKOU, Manager of the Economic and Social Statistics Division, and the overall leadership of Dr. Babatunde Samson OMOTOSHO, Director of the Statistics Department.

1. A case for innovation in data and statistics

1.1. Challenges facing official statistics in Africa and imperatives for innovation

Official statistics stand at a critical crossroads, confronted by challenges that test the very foundations of their relevance and resilience. Throughout history, they have demonstrated a remarkable capacity to adapt – navigating transformative shifts such as the rise of information technology in the mid-20th century – while steadfastly upholding fundamental principles that ensure their independence, integrity, and quality. However, the pressures of the 21st century represent an unparalleled convergence of threats, calling for a renewed commitment to innovation and institutional strength to safeguard the vital role of official statistics in shaping informed, evidence-based societies.

One of the foremost challenges confronting the statistical community today is the exponential surge in data generation. Every day, massive volumes of data are generated by economic, human, and societal activities, flooding an already saturated information ecosystem in which official statistics must compete for visibility and relevance. In Africa, significant investments in digitalization – such as numerous open data initiatives – have further increased data availability, compounding this growth. Yet, in this vast ocean of information, distinguishing high-quality, meaningful data from anecdotal or misleading content remains a significant challenge. As the classic economic adage warns, bad information often drives out good information. Information consumers are often impatient and want to have the most up-to-date data available as quickly as possible. Although readily accessible, some of these anecdotal data often originate from production processes that omit certain elements critical to ensuring their quality and representativeness. Official statistics, which are grounded in rigorous and complex methodologies that require time to ensure accuracy and reliability, are currently not positioned to compete in terms of timeliness.

For statisticians, this moment represents both a transformative opportunity and a call to evolve. Embracing big data is no longer optional, it is a paradigm shift that must be embedded in the fabric of statistical practice, enabling statisticians to remain relevant, authoritative, and impactful in a rapidly changing world.

The rapid proliferation of data, driven by the ubiquity of the internet and the rise of digital platforms such as social media, has fundamentally transformed how information is accessed and consumed. Today's data users are no longer passive recipients of information; they are discerning and proactive, seeking timely, granular insights tailored to their specific needs and interests. This shift has elevated expectations for relevance, responsiveness, and customization. In contrast, official statistics—often bound by rigorous but time-intensive production processes—face growing challenges in meeting these dynamic demands. To remain relevant, statistical systems must embrace innovative tools and approaches that align with the pace and precision of modern data consumption.

Technological advancements in data management are rapid and continuous, affecting not only ICT infrastructure but also the applications used for data collection, processing, storage,

analysis, and dissemination. Leveraging these tools effectively is essential for integrating vast data sources into the entire value chain of official statistics. However, keeping up with this rapid progress requires robust technological monitoring, organizational flexibility, and the capacity to mobilize and deploy the necessary resources to internalize new tools.

Equally important is the human factor, which plays a crucial role in driving innovation in statistical work. Harnessing new data sources and technological advancements requires skilled personnel

who understand the evolving landscape of their field. This necessitates awareness and training in various aspects of data science. The goal is not to turn every statistician into a data scientist but to ensure they comprehend the new statistical ecosystem and translate these insights into more efficient processes and methodologies.

Moreover, adopting emerging technologies in official statistics requires specialized expertise that is not always readily available within NSOs or across the broader National Statistical System (NSS). Public sector institutions often face significant constraints in attracting and retaining top talent, particularly when compared to the private sector, which typically offers greater flexibility and more competitive incentives. As a result, fostering a culture of innovation remains a critical challenge for many public institutions in Africa. Addressing this gap presents a powerful opportunity to strengthen institutional capacity and unlock the transformative potential of data-driven development.

These challenges emerge within a financial landscape marked by tightening constraints, underscoring the urgent need for a strategic and efficient allocation of limited resources to official statistics. Across Africa, national budgets face mounting pressure, and statistical systems often struggle to secure a place among top funding priorities. While international assistance remains a valuable source of support, it is increasingly limited, short-term, and narrowly focused. This reality hampers the pursuit of a long-term, transformative vision for data ecosystems – one that is essential for driving sustainable development and evidence-based policymaking

Considering these challenges, official statistics must evolve by adopting innovative approaches, leveraging technological advancements, and strengthening investments in human capital. Such a transformation is essential to ensure the continued delivery of reliable, high-quality data that underpins evidence-based policymaking and drives sustainable development.

1.2. Innovate or die!!!

Innovation is a crucial requirement for addressing the challenges associated with the ongoing transformation of data ecosystems in Africa. Data innovation for statistics involves the application of creative and advanced techniques in collecting, processing, analyzing, and presenting statistical information. This includes the use of cutting-edge technologies, methodologies, and practices to enhance data collection, processing, interpretation, and reporting. The goal is to improve efficiency, accuracy, and relevance across the entire statistical value chain. In an era of rapidly increasing data volume and complexity, embracing

innovation enables statisticians to uncover valuable insights, enhance data accuracy, and strengthen the impact of statistical analysis on decision-making and policymaking.

Transforming data ecosystems is an intricate and gradual journey. Yet, when innovation is strategically and thoughtfully applied, it becomes a powerful engine for progress across critical domains. Achieving sustainable impact through data innovation requires a well-grounded strategy – one that rests on robust infrastructure, strategic partnerships, skilled human capital, and meaningful community engagement. These elements are intimately intricated as shown in Figure 1.

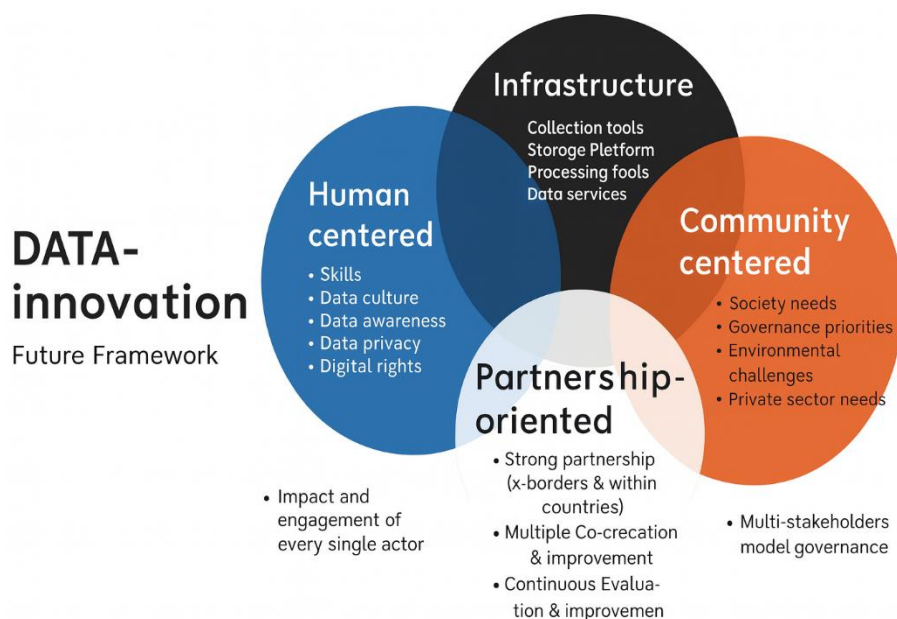


Figure 1: Building blocks for a data innovation strategy

An effective data innovation strategy should be structured around four pillars:

Infrastructure: Robust infrastructure forms the backbone of any effective data innovation strategy. It encompasses the physical and digital systems required to collect, store, process, and secure data on a scale. A well-designed infrastructure enables interoperability, supports advanced data pipelines, and accelerates experimentation; thus, allowing for efficiency in transforming raw data into actionable insights. As such, infrastructure underpins innovation, agility, and long-term competitiveness.

Partnerships: The challenges of transforming data ecosystems are too vast for NSOs to tackle alone, even though they are the most directly affected. Building innovative coalitions with partners in the NSS and other actors of the data ecosystem—some of whom may not have been traditional collaborators—is essential. Strategic partnerships are essential for amplifying the impact and reach of a data innovation strategy, providing access to specialized expertise, cutting edge tools, and complementary data sets. The sustainability of data-driven growth rests on partnerships that extend organizational capabilities and create a collaborative ecosystem.

Human Resources: Skilled and adaptable workforce are the driving force of any data

innovation strategy. Attracting, developing, and retaining talent with expertise in data science, analytics, artificial intelligence, and digital technologies ensures that the organization can effectively leverage its data assets. Thus, expanding capacity building initiatives is essential.

Societal Considerations: Integrating societal considerations into a data innovation strategy ensures that technological advancements create inclusive, ethical, and sustainable impact. Transforming data ecosystems through innovation requires rethinking data governance and its priorities. This pillar encompasses data privacy, security, ethical use of AI, and compliance with regulatory frameworks. It also entails far-reaching strategic and institutional reforms to meet society's evolving information needs.

1.3. Modernisation versus transformation

Most of the National Strategies for the Development of Statistics (NSDS) that have been prepared in recent years by African countries focus on the modernization of the national statistical system. They include interventions to be undertaken in the areas of ICT equipment and infrastructure, human resource capacity building, and reform of the legal and institutional framework for the national statistical system. On institutional framework, the emphasis is on strengthening partnerships among members of the NSS for impactful outcomes. Some strategies emphasize the use of new data sources for statistics, in particular the more intensive use of administrative data or «big data». Innovation is often mentioned but as a specific activity, detached from the rest of the proposals.

Given the complex and evolving challenges facing official statistics today, it is imperative for National Statistical Offices to move beyond the scope of modernization outlined in their current NSDSs. A more transformative and forward-looking approach is required - one that not only retools statistical systems but also reimagines their role in driving evidence-based development. Transitioning from the modernization of the NSS to the transformation of National data ecosystems for greater impact requires improvements in three key areas:

- First, the challenges confronting official statistics involve actors who are outside the NSS in the traditional sense. Therefore, it is important to conceptualise data innovation within the broader context of the national data ecosystem, encompassing all organizations and structures that produce and use data within a country, rather than focus solely on the NSO and its partners within the NSS. These new actors are not primarily tasked with producing official statistics, as members of the NSS are. However, they collect data – either directly or as a by-product of their activities – that can be valuable for constructing and/or supplementing official statistics. These actors must be engaged through a coordinated global approach designed to transform the approaches, tools and processes that make it possible to produce official statistics.
- Second, innovation must be integrated into statistical operations in a horizontal manner and should not be considered separately. It constitutes an integral component of a coherent, overarching strategy for transforming the ecosystem, playing a vital role at every stage of the statistical value chain and for all the stakeholders involved.

- Third, it is essential to give a prominent place to coordination and collaboration between the actors of the data ecosystem in terms of innovation. This must be done comprehensively and not just around the exchange of data, as is often the case currently. It is also important to not only create coalitions that make it possible to share experiences, both good and bad, in terms of innovation, but also to work together in creating and deploying innovative solutions throughout the value chain of statistical production.

Table 1 below presents a rough synthesis of the main differences between modernizing the NSS and transforming the data ecosystem.

Table 1: Modernisation versus transformation

	Modernisation of the statistical system	Transformation of the data ecosystem
Scope	Improve quality, reliability, availability and accessibility of (official) statistics	Create a dynamic, open and collaborative environment around data.
Nature of the process	Internal, institutional	Global, systemic
Who is concerned	Mainly the members of the NSS	Beyond the NSS, all owners, producers and users of data
Main related issues	Reliability, independence, accessibility of official statistics	Inter-operability, transparency, governance, innovation

2. Innovation in the value chain of official statistics

Innovation must be considered at every stage of the statistical value chain. However, to be truly effective, innovation must originate at the grassroots level, driven by the technicians responsible for operational processes. These professionals have the deepest understanding of the tools they use and the workflows they follow. As such, they should play a central role in any efforts to improve or transform these tools and processes. Ideally, they should be empowered to initiate innovation themselves. This requires providing them with the necessary skills, time, and support to critically assess their work and explore ways to enhance efficiency.

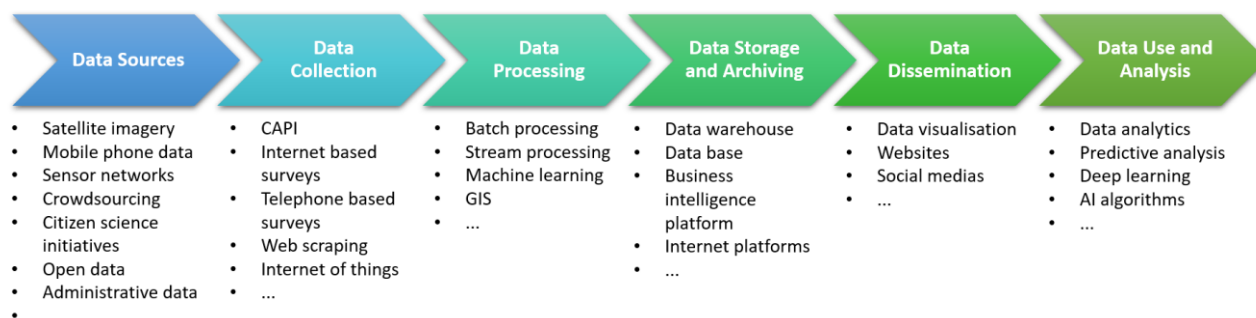
While certain aspects of innovation may require specialized expertise and targeted technical support, those who implement statistical processes must remain at the forefront of applying and adapting innovations. Their close involvement ensures that changes are practical, relevant, and sustainable. Ultimately, innovation holds value only when it addresses real needs and is clearly understood by those responsible for implementing it.

The statistical value chain typically consists of six key stages: **data mobilization, collection, processing, storage and archiving, dissemination, and analysis**. These stages¹ are interdependent, often overlap, and continuously evolving. Figure 2 below

¹ Methods and definitions and standardization are also important elements in the first stage of the data value chain coming before collection. These elements are not considered here but they may also benefit from

provides an illustrative, though non-exhaustive, overview of potential innovations at each stage.

Figure 2: Innovation in the statistical value chain



- a) **Data mobilization** – The data mobilization stage involves the systematic gathering of information from various sources, some very traditional to statistical work, including censuses, surveys and administrative records, and other less conventional such as sensors, digital platforms, satellite imagery or mobile phone records, among others. Mobilizing these new sources is today a foundational stage of statistical work, where significant advancements have been made in the recent past.
- b) **Data collection** – Innovations include improvements in traditional statistical operations (such as surveys and censuses) and the integration of alternative data sources (such as administrative records and non-traditional data streams). Advances such as mobile data collection tools, computer-assisted interviewing, web scraping, satellite imagery, and the integration of big data sources have enhanced timeliness, accuracy, and cost-efficiency of data collection. Also, these advances have expanded the range of phenomena that can be measured, improving the relevance and responsiveness of official statistics.
- c) **Data processing** – This refers to the cleaning, validation, integration, and transformation of raw data into structured, reliable datasets suitable for analysis. This stage has benefited significantly from advancements in computing and automation, including the adoption of automated data validation systems, machine learning algorithms for data cleaning, and the use of metadata-driven workflows to ensure consistency. Transforming raw data into reliable and high-quality statistical products involves numerous operations, all of which must ensure accuracy, consistency, and reliability. Cloud-based processing platforms and data pipelines now enable faster, more scalable, and more collaborative processing, reducing manual effort and minimizing human error.
- d) **Data storage and archiving** – This stage focuses on the secure, efficient, and sustainable preservation of data and metadata for both current use and future reference. The increasing availability of data, coupled with growing demand for both historical and granular information, places considerable pressure on storage and

innovation.

archiving processes. Innovations have emerged through the adoption of cloud storage solutions, distributed ledger technologies, and advanced database systems that support large-scale, high-frequency data. The use of standardized data repositories, data catalogues, and preservation protocols ensures that statistical data remains accessible, interoperable, and trustworthy over time. Enhanced cybersecurity and data governance frameworks have also strengthened the integrity and confidentiality of stored information.

- e) **Data dissemination** – Data dissemination involves making statistical information available to users through accessible, user-friendly channels. This stage is undergoing profound transformation, with open data portals, interactive dashboards, APIs, and data visualization tools now standard practice in many national statistical systems. These innovations promote transparency, encourage reuse, and support evidence-based decision-making by allowing users to explore, download, and visualize data dynamically. The move toward open data standards and interoperability has further facilitated integration across institutions and domains. Given the diverse needs of the users, dissemination strategies must be flexible, adaptable, and responsive to various audiences and requirements.
- f) **Data analysis** – The ultimate purpose of statistical production is to generate insights that support decision-making, democratic debate, and policy formulation. Data analysis is the stage where insights are derived through statistical methods, econometric modelling, and increasingly, artificial intelligence and machine learning techniques. Analytical innovations not only enhance the interpretation of statistical data but also play a critical role in assessing data quality and improving future statistical processes. Innovations in this domain have broadened analytical capabilities beyond traditional statistical methods to include predictive analytics, real-time monitoring, and complex network analysis. The use of advanced visualization tools, programming languages like Python and R, and collaborative analytics platforms has empowered analysts to generate deeper, faster, and more actionable insights — ultimately improving the quality and relevance of statistical outputs for policy and development planning.

Given the dynamic nature of the statistical landscape, continuous innovation is essential to ensure that statistics, and especially official statistics, remain relevant, reliable, and responsive to societal needs.

3. What is the statistical community doing on innovation?

Numerous initiatives are currently driving the transformation of the African data ecosystem – both through broad, cross-cutting efforts and sector-specific interventions. These initiatives address not only overall innovation in statistics but also targeted areas such as data governance, the development of standards and norms, and the responsible use of generative AI. The sections that follow highlight some of these initiatives. As innovation gains increasing attention within the statistical community, the number and scope of such efforts are expected to

expand significantly in this critical domain.

3.1. United Nations Commission on Statistics and Data²

The Commission's general framework for intervention is the «Global Digital Compact» which aims to:

- Eliminate all disparities in digitalization;
- Advance accountable, equitable and interoperable approaches to data governance; and
- Strengthen international governance in artificial intelligence.

The Commission's agenda has 4 elements:

- **Data governance:** This focuses on advancing data governance to enhance accountability and inclusiveness within data processes and institutions. It entails clearly defining the roles and responsibilities of all stakeholders involved to ensure coordinated, transparent, and equitable management of data resources.
- **Partnership on ICT:** This is a global initiative which aims to improve the availability of high-quality and internationally comparable statistics in the ICT sector.
- **The Committee of Experts on Big Data and Data Science for Official Statistics (UN-CEB- D):** The committee was set up following the 45th meeting of the Statistical Commission and comprises 9 teams working on various aspects of the topic namely Automatic Identification System data, big data and SDGs, Earth observation data, rural access measurement, mobile phone data, privacy-enhancing technologies, scanner data, training, skills and capacity development, and access to private data. It has established an ecosystem of cloud services to support international collaboration in the development of official statistics using new data sources and innovative methods, and to help countries monitor the progress towards the Sustainable Development Goals (SDGs).
- **International events.** This element provides a platform for extensive exchange of views among participants from diverse backgrounds on critical issues in data science. For example, the session on “Artificial Intelligence and Data Science for Economic Statistics” held in 2024 exemplified this collaborative engagement.

3.2. “Data for Now” (UNSD)³

The objective of the initiative is to increase the use of robust and innovative methods and tools to improve the timeliness, coverage and quality of data on SDGs and other key national development indicators. It also aims to ensure that the new data and statistics can be easily accessible and used to inform policymakers, stakeholders and development partners to achieve the 2030 Agenda for Sustainable Development.

² <https://www.un.org/digital-emerging-technologies/global-digital-compact>

³ <https://unstats.un.org/UNSDWebsite/capacity-development/data-for-now/>

3.3. Roadmap for the transformation and modernization of official statistics in Africa, 2023–2030 (UN-ECA)⁴

The rationale for this work is the great diversity that exists between national statistical systems and national data ecosystems, as well as the variety of resources and policy contexts to be considered in African countries (Figure 3). The transformation and modernization of official statistics, including approaches to digitalization, requires national strategies that are perfectly adapted to national and often subregional circumstances.

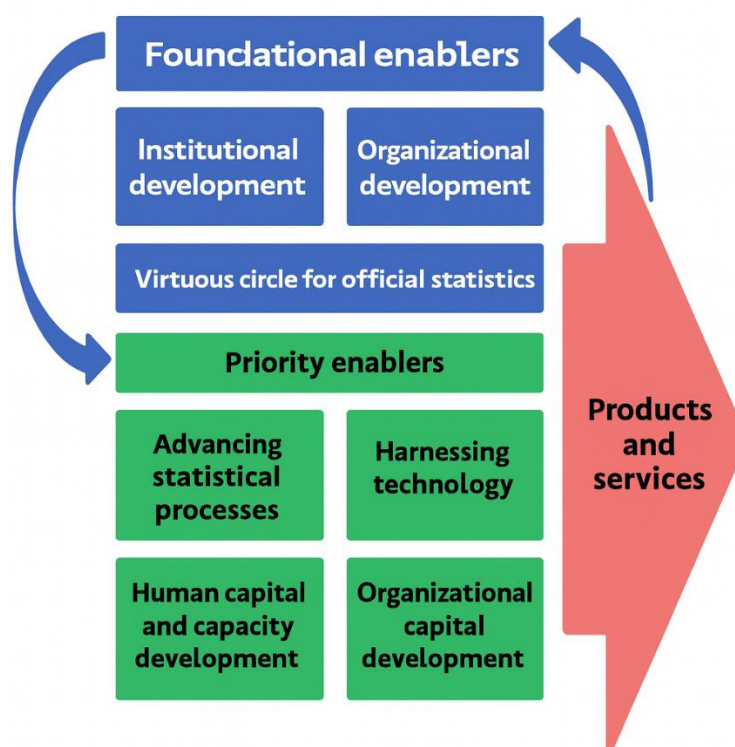


Figure 3: Recast foundational and priority enablers for transformation and modernization

Source: Strategic toolkit for the transformation and modernization of national statistical systems in Africa, United Nations, Economic and Social Council, 23 September 2024

Other technological developments, basic physical infrastructure, human resources (including technical and managerial capacities), business process design and professional support networks are also considered in the approach developed by UN-ECA.

3.4. Data Innovation Fund (World Bank)⁵

The World Bank was particularly interested in projects that produce data or methods faster, at lower cost and/or with higher resolution or granularity, to fill critical data gaps. Ultimately, the goal of the DIF was to encourage collaboration, experimentation, learning, and capacity development in the field of sustainable development data. DIF closed its doors at the beginning of 2021.

The Data Innovation Fund (DIF) funded scalable, repeatable, and innovative approaches to

⁴ <https://repository.uneca.org/handle/10855/49338>

⁵ <https://www.worldbank.org/en/data/statistical-capacity-building/data-innovation-fund>

improve data in the field, with a focus on knowledge sharing. During three calls for proposals in 2016, 2017 and 2018, the Fund received around 900 project proposals in support of several SDGs in sub-Saharan Africa, East Asia and the Pacific, Europe and Central Asia, Latin America and the Caribbean, and the Middle East and North Africa, and South Asia. The 38 winning projects received funding ranging from \$25,000 to \$250,000 each.

3.5. Strategy for digital transformation in Africa 2020-2030 (African Union)⁶

This strategy initially focused on the formulation of an African Union convention on cybersecurity and data protection, areas that were not covered by this type of continental agreement but where there was a dire need for a comprehensive and concerted approach (Figure 4).

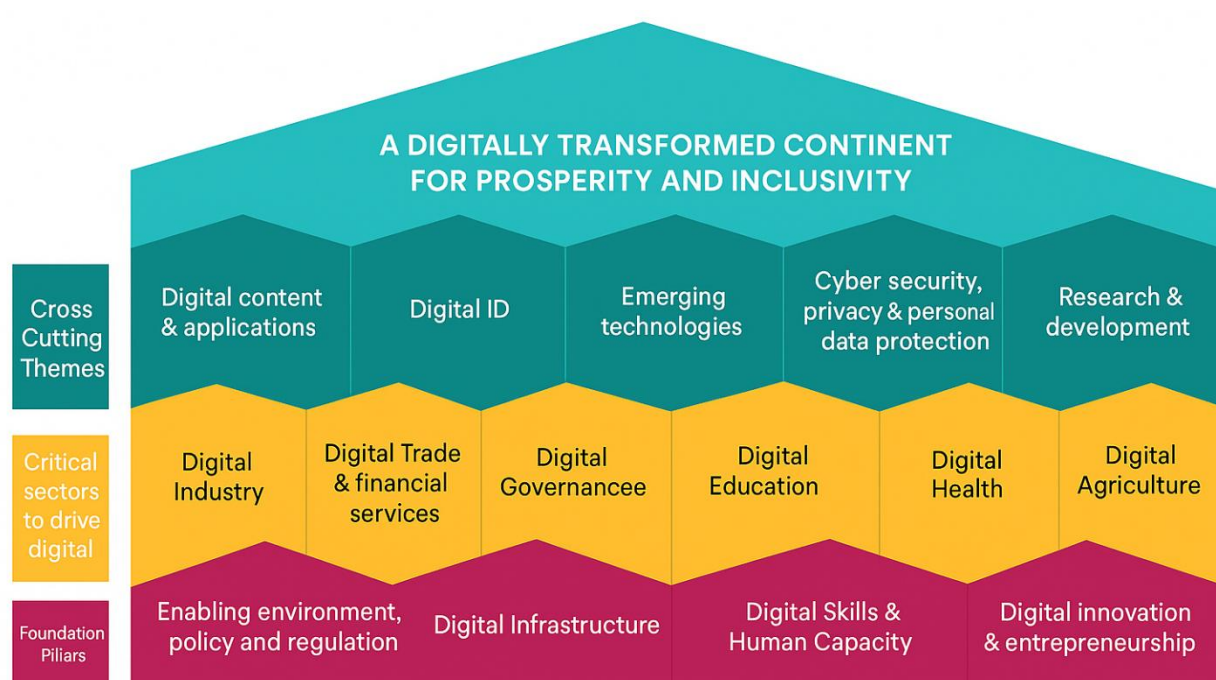


Figure 4: The digital transformation strategy for Africa (2020–2030)⁷

The strategy aims to «develop inclusive digital skills and human capacities in the field of digital sciences and education, both technical and professional, to lead and fuel digital transformation, including coding, programming, analytics, security, blockchain, machine learning, artificial intelligence, robotics, engineering, innovation, entrepreneurship, and technology policy and regulation. »

4. The AfDB's Data Innovation Lab initiative

4.1. What is the AfDB DIL?

The African Development Bank's Data Innovation Lab is a strategic initiative designed to

⁶ <https://au.int/en/documents/20200518/digital-transformation-strategy-africa-2020-2030>

⁷ Source: African Union

foster data-driven innovation, enhance statistical capacity, and accelerate the digital transformation of data ecosystems across Africa. Anchored on two core pillars – creating innovation for data value and impact and building networks and collaboration across the data ecosystem – the DIL serves as both an incubator and a collaborative platform for co-creating, testing, scaling, and institutionalizing innovative approaches to data collection, management, and analysis. the DIL aims to strengthen the Bank’s and Regional Member Countries’ ability to generate high-quality, timely, and policy-relevant data. Ultimately, the Lab contributes to the creation of an inclusive, interconnected, and sustainable data ecosystem that empowers evidence-based decision-making and supports Africa’s development agenda.

4.2. AfDB DIL objectives

The AfDB’s DIL initiative has 4 interrelated objectives:

Objective 1: Enable innovation in data and statistics within the Bank and Regional Member Countries (RMCs). The DIL seeks to create an **enabling environment for innovation in data and statistics**, both within the African Development Bank and across its Regional Member Countries. This involves promoting a culture of experimentation, collaboration, and creative problem-solving in the production and use of data. Through capacity-building, pilot projects, and partnerships with academia, the private sector, and civil society, the DIL will empower teams to explore new data sources, methodologies, and technologies. By institutionalizing innovation practices, the DIL ensures that data-driven approaches become integral to policy design, operational efficiency, and development impact across the continent.

Objective 2: Produce augmented and cost-effective data and statistics for better decision-making outcomes. The DIL aims to **enhance the quality, timeliness, and affordability of data production** using advanced analytics, automation, and emerging data technologies. By integrating traditional statistical methods with innovative data sources such as satellite imagery, geospatial data, and administrative systems, the DIL will help generate richer and more policy-relevant datasets. These innovations will support evidence-based decision-making within the Bank and among African policymakers, improving development planning, monitoring, and evaluation. Furthermore, by demonstrating scalable and cost-effective models, the DIL contributes to reducing the data gaps that hinder inclusive and sustainable growth on the continent.

Objective 3: Demystify big data and data science by developing skills within the Bank and its RMCs. A core objective of the DIL is to **demystify big data and data science, making these tools more accessible and understandable** to statisticians, policymakers, and development practitioners. Through hands-on training, workshops, and collaborative projects, the DIL will enhance data literacy and technical expertise across the Bank and RMCs. The initiative seeks to bridge the gap between data specialists and decision-makers by promoting a common understanding of how advanced analytics, artificial intelligence, and machine learning can be responsibly applied in development contexts. By building institutional and human capacity, the

DIL fosters confidence and competence in using data science to address Africa's most pressing challenges.

Objective 4: Build a repository of tested and free open-source tools for broad use by the Bank and its RMCs. The DIL aims to **develop and maintain a repository of tested, open-source tools and methodologies** that can be freely accessed and adapted by the Bank and its member countries. This repository will include data management frameworks, analytical models, visualization applications, and automated workflows designed to address common challenges in data collection, processing, and analysis. By promoting openness, interoperability, and reusability, the DIL encourages collaboration, reduces duplication of effort, and accelerates the adoption of proven solutions. Ultimately, this objective supports the broader goal of building a robust, inclusive, and sustainable data ecosystem for Africa's development.

4.3. AfDB DIL implementation pillars

The two main pillars underpinning the AfDB's Data Innovation Lab objectives are as follows:

Pillar 1: Creating innovation for data value and impact. This pillar focuses on **fostering user-driven innovation** that enhances data access, quality, and utilization across the African data ecosystem. It emphasizes designing solutions that respond directly to user needs—whether within the Bank, among Regional Member Countries (RMCs), or across broader development communities. The DIL will accelerate the innovation cycle by supporting the incubation, testing, and scaling of data-driven initiatives once they are conceived. Central to this effort is the augmentation and responsible use of diverse data sources, including administrative, geospatial, and big data, to generate actionable insights for policy and investment decisions.

Also, this pillar prioritizes the **development of data skills and literacy** among professionals and decision-makers. Through structured capacity-building programs, mentorship, and knowledge-sharing, the DIL will cultivate a new generation of data innovators equipped to harness emerging technologies such as artificial intelligence, machine learning, and advanced analytics. Moreover, the pillar underscores the importance of **strong leadership and an enabling legal and regulatory environment** for data governance, ensuring that innovation proceeds within ethical, secure, and transparent frameworks. Together, these actions will create a foundation for sustained, inclusive, and scalable data innovation.

Pillar 2: Building networks and collaboration across the data ecosystem. The second pillar aims to **strengthen collaboration and connectivity among data innovators** by building a vibrant, inclusive, and interoperable ecosystem. The DIL will serve as a platform for networking and co-creation among key actors — including national statistical offices, government agencies, research institutions, private sector innovators, civil society organizations, and development partners. Through collaborative environments, the Lab will promote the **sharing of knowledge, methodologies, and data resources**, enabling participants to learn from each other's experiences and align their efforts toward common priorities.

This pillar also focuses on **facilitating technology transfer and fostering complementarity** and interoperability among systems and institutions. By promoting open standards, shared platforms, and data exchange protocols, the DIL will enhance efficiency and reduce duplication across the ecosystem. Furthermore, the Lab will play a **catalytic role in mobilizing funding and partnerships** to support innovation, encouraging co-investment in transformative data initiatives. Ultimately, this pillar seeks to build a connected, collaborative, and sustainable community of practice that accelerates data innovation for Africa's development.

4.4. AfDB DIL principles

The AfDB Data Innovation Lab (DIL) operates under a set of guiding principles that reflect the Bank's commitment to transforming Africa's data landscape into an ecosystem of innovation, collaboration, and sustainability. These principles—illustrated in *Figure 5*—define how the DIL approaches data, partnerships, and innovation in practice.



Figure 5: Principles of the AfDB DIL initiative

a) Data augmentation

The DIL champions the augmentation of traditional statistical systems by integrating non-traditional, high-frequency, and alternative data sources such as satellite imagery, mobile phone records, fintech transactions, and sensor-generated data.

This principle ensures that national and regional institutions move beyond periodic surveys toward real-time, granular intelligence capable of capturing the dynamism of Africa's economies and societies. Through data augmentation, the DIL helps unlock new insights to support the Bank's High 5s, SDGs, and Agenda 2063—strengthening the analytical foundations for evidence-based policy, planning, and service delivery.

b) Collect once, use and reuse

Efficiency and sustainability are at the heart of this principle. The DIL promotes interoperable

systems and data architectures that allow information to be collected once, stored securely, and reused across multiple purposes, institutions, and sectors. By reducing duplication, transaction costs, and data loss, this approach fosters a culture of standardization, documentation, and long-term value creation.

c) Reduce friction

Data innovation can only thrive when institutional, technical, and procedural barriers are minimized. The DIL therefore focuses on reducing friction throughout the data value chain—streamlining access, standardizing formats, and promoting interoperability across systems. This includes aligning with open standards such as SDMX, deploying automation tools for data integration, and establishing shared platforms for storage and dissemination. Reducing friction ensures that data moves seamlessly from collection to decision-making, enabling faster, more reliable insights that directly inform operations and policy.

d) Partnerships as a core value

Beyond being a principle, partnership is the defining ethos of the DIL. The Lab's success depends on collective intelligence, shared ownership, and co-creation across stakeholders. By cultivating trust-based relationships and shared governance frameworks, the DIL transforms collaboration into a strategic asset—bridging silos within the Bank and across Regional Member Countries (RMCs). This cooperative model ensures that Africa's data transformation is inclusive, sustainable, and anchored in continental leadership.

4.5. AfDB DIL workstreams

In practice, the DIL initiative has been structured around four complementary workstreams that translate its strategic vision into tangible results: *(1) innovation and applied use cases, (2) capacity building and knowledge transfer, (3) data ecosystem diagnostics and networking, and (4) strategic partnerships and governance for data innovation.*

Through these workstreams, the Lab promotes data-driven innovation, strengthens institutional and human capacities, and fosters collaboration across Africa's data ecosystem.

Appendix 7 presents a comprehensive overview of the key activities carried out between 2022 and 2025, showing how each contributes to the DIL's overall framework by aligning with its objectives, implementation pillars, and guiding principles.

5. The DIL initiative in the context of AfDB's support to the African Statistical System

The AfDB has, for many years, supported African states in strengthening their statistical systems and promoting a culture of evidence-based decision-making. The Bank's overarching goal in supporting the African Statistical System can be summarized in the phrase: **"Reveal the Power of Data in Africa"**.

This support gained new momentum in 2012, with the launch of the Africa Information Highway programme, which focused on:

1. Facilitating public access to official statistics,

2. Reducing reporting burden in statistical operations,
3. Promoting the use of international norms and standards,
4. Strengthening openness and transparency in statistical systems,
5. Strengthening statistical ecosystems.

The AIH is set up around 4 main components:

1. Orchestration of the continent's data ecosystem,
2. Implementation of «open data» portals, at the regional and in-country levels,
3. Participation in and promotion of the « Statistical Data and Metadata Exchange - SDMX»,
4. Support to data innovation.

These three components are presented in Figure 6.

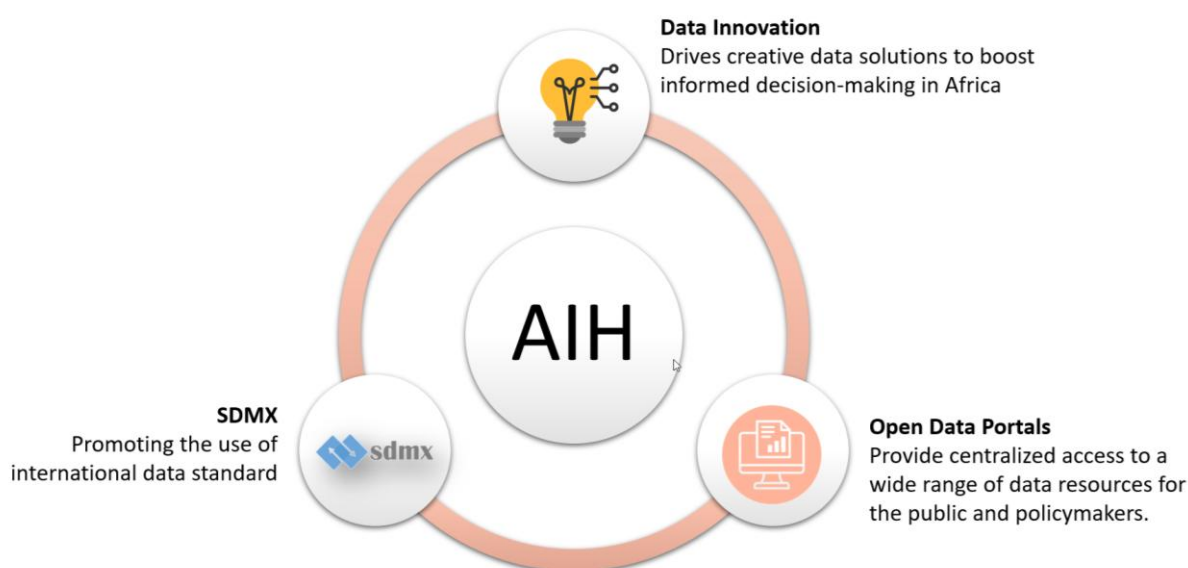


Figure 6: Components of the AfDB's AIH

As part of the data innovation component of the AIH programme, the Bank joined the Development Data Partnership initiative in 2023, with the aim of:

- a) Facilitate third-party access to innovative data on Africa,
- b) Initiate and support collaborative alliances and advanced projects on the development of the continent,
- c) Deploy distance learning methods for African countries.

The AfDB's overall interventions under this new component have featured a pilot project on measuring the accessibility of health sector services using various available data sources (granular population data, geo-data and hospital information).

6. What is a Data Innovation Lab?

6.1. Main functions

A DIL is basically a structured platform designed to foster cooperation, experimentation, and exchange around the use of data and technology for development. It brings together data

producers, users, and innovators from diverse sectors – such as government institutions, academia, the private sector, and development organizations - to co-create solutions that address complex data challenges.

As illustrated in Figure 7, a DIL must perform three interconnected roles as follows:

- *DIL as a catalyst.* It must be the meeting point for cooperation and knowledge exchange on innovation in statistics. A DIL plays a pivotal role in accelerating innovation and transforming how data is used for development. It stimulates new ideas, encourages experimentation, and promotes the adoption of emerging technologies and methodologies across institutions. By connecting diverse actors – data producers, users, and innovators – the DIL creates synergies that help bridge gaps between research, policy, and implementation. It challenges traditional approaches to data management and analysis, inspiring new ways of thinking about data generation, sharing, and application. Through targeted pilot projects, thought leadership, and advocacy, the DIL helps embed a culture of innovation within organizations and across the wider data ecosystem, ultimately driving systemic change.
- *DIL as a facilitator/enabler.* It must be able to provide a collaborative environment where stakeholders can share knowledge, exchange best practices, and co-develop solutions to common data challenges. It fosters partnerships between government agencies, private sector entities, academia, and development institutions, ensuring that innovation is both inclusive and contextually relevant. A DIL also enables capacity development by equipping data practitioners and policymakers with the technical and analytical skills needed to leverage advanced data tools and methods. Moreover, it supports the creation of enabling conditions - such as governance frameworks, interoperability standards, and data-sharing protocols – that ensure sustainability, trust, and ethical data use. A DIL also supports financing of data innovation experiments and the establishment of coalitions and partnerships facilitating the pooling of resources and means.
- *DIL as an innovator.* A DIL functions as a testbed for the design, development, and deployment of cutting-edge data solutions. It explores new frontiers in data science, such as artificial intelligence, machine learning, geospatial analytics, and real-time data integration, to improve the efficiency and effectiveness of data systems. The Lab enables experiments with open-source tools, automation, and novel data sources (including big data and administrative records) to enhance the scope and quality of statistical outputs. Beyond technology, the DIL also innovates in processes - introducing agile approaches to project management, rapid prototyping, and iterative learning. By demonstrating practical, scalable applications of innovation, the DIL not only generates proof-of-concept models but also provides replicable frameworks that can be adopted widely. Engagement in this role must not be autonomous, at the risk of becoming isolated, but must be dictated by a collaborative approach based on the expression of needs by users and experts in the value chain of the data ecosystem.

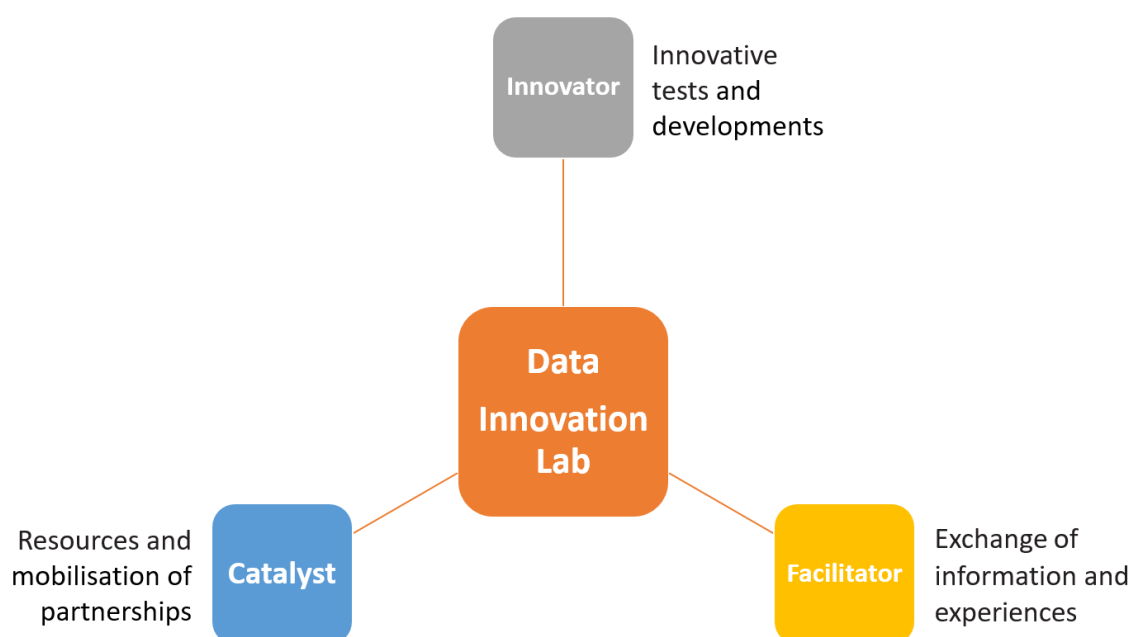


Figure 7: Functions of a DIL

6.2. Establishing a DIL

The establishment of a DIL is a complex process that goes beyond its simple physical or virtual creation. The process must be gradual, both to ensure broad support for its role/mandate within the national statistical offices but also with partners within the NSS and beyond. It must develop activities that will be of various kinds and that will target different timelines, in the short, medium and long term. There is a need to build a clear and consensual vision on the importance of innovation for official statistics in the NSO and in the entire data ecosystem, on the role that the NSO can and should play in this field and on the need to exchange and share experiences across the entire value chain within the national data ecosystem. See also the section “*A physical structure or a virtual platform to host the DIL?*” below for more details on the establishment of a DIL.

6.3. The AfDB’s DIL diagnostic model

Figure 8 illustrates the maturity model which is used for the AfDB-DIL diagnosis is built on an assessment that covers 12 dimensions. These dimensions are complementary, and they converge to analysing the necessary conditions that must be met to effectively develop a relevant and sustainable path towards the transformation of the data ecosystem. These 12 dimensions are represented in the diagram below.

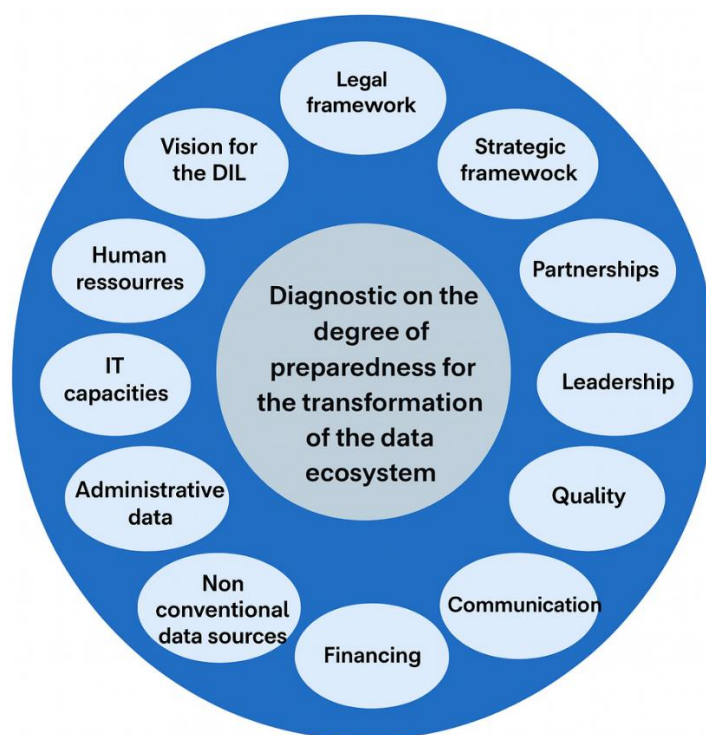


Figure 8: The 12 dimensions of the maturity model

The AfDB-DIL diagnostic model does not assign relative importance of each dimension, nor is it intended for grading or classification. Its primary purpose is to support the establishment and operationalization of DIL activities. The diagnostic framework serves as a practical tool for conducting an initial situational assessment, while also providing a mechanism for ongoing monitoring and progress tracking. The initial assessment is carried out collaboratively with the AfDB team to ensure clarity on concepts and content; thereafter, the framework is designed to be used autonomously by the participating countries, enabling them to independently evaluate and enhance their DIL initiatives overtime.

The 12 dimensions are described briefly below.

Dimension 1: Leadership and «benevolent» support. This dimension looks at the extent to which the justification for setting up a Data Innovation Lab is well rooted and understood by NSO management and staff. It is divided into two main parts: one dealing more with practices for disseminating innovation and resistance to change within the NSO, and the other focusing on the transformative vision, its documentation and sharing.

Dimension 2: Strategic vision of the transformation. The aim here is to gain a better understanding of the level of understanding and awareness within NSO, but also more widely in the country, of the transformation underway generated by the rapid increase in digitalisation and data availability (Open data) but also by the rapid technological progress in the field (AI in particular). It is assessed via four main topics: statistical strategy (is there a strategy for innovation and what its content is in terms of data management), statistics in national policies and strategies, data and/or AI strategy, and organization and coordination within the NSO.

Dimension 3: Legal and institutional framework. The aim is to assess the room for

maneuver that exists for the NSO to change (make evolve) what is established in terms of responsibility and coordination within the data ecosystem (beyond the simple NSS and the statistical law in force). This dimension is presented in six main components: coordination within the NSS, exchange of experience within the NSS, the data stewardship function of the NSO, interconnectivity among the actors of the data ecosystem, the role of the National Statistical Council if there is one, and the way the SDGs are monitored.

Dimension 4: Human resources. The aim is to identify how to make human resources an effective force/asset in implementing activities aiming at the transformation of the data ecosystem. This involves a review of what exists at the NSO and beyond. This dimension is divided into 3 main components: data science skills, capacity building schemes, and partnerships for training in data science.

Dimension 5: Information technology. Is the technological potential existing at NSO sufficient to initiate and deepen the transformation by consolidating the innovative experiences and developments? This requires a benchmark covering of IT equipment and complex applications linked to the use of AI tools, for example. This dimension is broken down into 3 main components: IT infrastructures for data collection, AI applications, and analysis tools.

Dimension 6: Administrative data. This dimension looks at the extent to which the NSO has invested in the mobilisation of administrative data for statistical purposes and what good practices exist in this area. This dimension is presented in 3 main components: the state of play, current experiences and problems encountered.

Dimension 7: Non-traditional sources. The aim is to find out the extent to which the NSO has invested in mobilising non-statistical (non-conventional) data for statistical work and whether there are any good practices to be identified and replicated in this area. This dimension is divided into 2 main components: experiments in progress and those relating more specifically to the use of mobile phone data.

Dimension 8: Partnerships. This dimension seeks to assess which collaborations and coalitions can be used to share investment in innovation and disseminate the results of transformation efforts at the NSO within the NSS and beyond. This dimension is divided into 2 main components: partnerships within the NSS, platforms and partnerships outside the NSS.

Dimension 9: Quality, norms and standards. This dimension looks at the extent to which the NSO applies regional and international technical recommendations on quality and standards, and the extent to which these recommendations are also integrated into the work of other stakeholders in the country. Is this notion of quality a clear foundation of the reputation of the NSO? What does the NSO do to monitor the quality of its work. This dimension is divided into 6 main components: the quality models used, quality culture, exchanges between the NSO and its partners on these issues, the norms and standards used in the NSS, the support given to the NSS by the NSO and the reputation of the NSO among its partners in these areas.

Dimension 10: Communication and relations with users. The aim is to identify how the trust factor can promote transformation through the recognition of the quality of the NSO's work and its impact. This dimension is divided into 6 main components: NSO's communication policy, its relations with users, the responsiveness and agility of the structure, the availability of data, treatment given to specific requests for data and monitoring of the use of data in the country.

Dimension 11: Research and analysis expertise. The aim is to find out what level of advanced analysis has been achieved by the NSO. It also seeks to assess the availability and use of data for analysis in the country in the academic and research world. This dimension is divided into 5 main parts: collaboration of the NSO with the academic world, use of artificial intelligence, existence of complex analyses, access to micro-data and research potential in the country.

Dimension 12: Technical and financial support. The aim is to find out what national and external funding opportunities might be available to support the DIL's activities. This dimension is divided into 2 main components: the existence of a group of donors and future projects to support statistical activity and the support received by statistics from the national budget.

Detailed questions for each dimension are presented in Appendix 1. It should be noted that these are open-ended questions and therefore there is a margin of flexibility to adapt them to the context of each country.

6.4. Who should perform the diagnostic?

It is useful to specify that the diagnosis is carried out to support the vision and the approach that the National Statistical Office of each country has initiated to develop and consolidate its efforts and investments towards innovation. It is thus primarily intended to accompany the ongoing national process, to generate more thoughts on approaches for partnership and the use of non-conventional data sources and to support the foundations for the establishment of a DIL and the deployment of its activities.

The maturity model associated with the DIL is not intended as a benchmarking tool for comparing African countries' innovation in statistics. Instead, it serves as a framework to help stakeholders of the national data ecosystem assess their readiness to integrate innovation into their work. By providing a snapshot of their current context and practices, the model supports informed decision-making for a future aligned with digital transformation and technological advancements.

Ideally, African states should conduct this assessment themselves, with the NSO playing a central role as coordinators of the NSS, with the involvement of key actors in the data ecosystem. External expertise can also be leveraged to provide a more neutral perspective. However, these assessments were carried out in close collaboration with the NSOs of the participating countries.

Based on experience, a mixed approach could be beneficial, combining self-assessment by

states with targeted external expertise for specific aspects of the diagnosis. Alternatively, an initial assessment supported by external experts could lay the foundation for subsequent evaluations conducted by local actors, drawing on the knowledge and insights gained from the first assessment.

Regardless of the approach taken, the most critical aspect of the diagnostic process is fostering dialogue among a broad range of stakeholders within the data ecosystem while also mobilizing expertise within the NSO. This process provides a valuable opportunity to share the NSO's vision for innovation, address data management and governance challenges, and build collective commitment to progress. Some areas of the maturity model extend beyond the NSO's scope and require collaboration with NSS partners and other key players in the data ecosystem. Therefore, a cooperative and inclusive approach is essential for meaningful development.

6.5. How to best present the results of the diagnostic?

Once the diagnosis has been established, it is essential to present the results in a way that makes them easily interpretable—primarily through visual representation. This is why the results are conveyed using a color-coded system for each individual question:

- **Green:** The situation is favourable.
- **Orange:** There is room for improvement and sustained attention is required.
- **Red:** The situation demands further reflection and the implementation of priority measures.



The objective is not to assign blame or rank and compare, but rather to evaluate the situation as objectively as possible. This assessment provides a critical foundation for reflection, informing strategic decision-making and guiding the effective design and implementation of the **DIL**. Importantly, the diagnostic process itself—along with the critical thinking it fosters—is more valuable than the mere assignment of colour codes. At this stage, the focus should be on identifying general trends rather than rigidly defining criteria for assigning colours. As a reminder, the goal is neither to judge nor to compare but to provide strategic direction for investment and innovation efforts.

Once each question has been colour-coded, the diagnostic results are presented in two distinct formats, each serving a different purpose:

1. **Technical analysis for DIL deployment.** A detailed assessment is conducted by responding to each question in the diagnostic framework. If a question is not relevant to a specific country's context, it should be adapted or replaced accordingly. In some cases, presenting multiple perspectives (i.e., different responses) to a single question can offer deeper insights. For diagnostics conducted with the support of the AfDB's DIL initiative, test countries provided initial responses, which were then reviewed and supplemented with external expertise.
2. **Sharing and promoting results.** A summarized version of the findings—illustrated in a summary table (see Appendix 2)—provides an accessible overview of

the results. This summary can be used to engage partners in the ecosystem, decision-makers, and donors. The level of detail can be adjusted based on the audience, whether through a high-level overview (12 colour-coded dimensions) or a more granular breakdown (approximately 60 colour-coded components).

7. What typical interventions can be deployed through a national DIL project?

7.1. Supporting versus steering factors

Proposals for the installation of a DIL must consider various factors that are covered by the questionnaire. These are categorized as follows:

- a) Supporting factors: these are elements related to the NSO's commitment to a transformative vision of the data ecosystem, its resources in people trained in data science and IT infrastructure, the stock of innovative experiences already carried out across the entire value chain of statistics and the attention given to the quality and the application of international norms and standards,
- b) Steering factors: these are elements over which the NSO has less direct control but which can be decisive in advancing the transformative agenda such as the strategic, institutional, legal and political frameworks for data management, the existence and intensity of partnerships within the ecosystem on innovation in data management, the opening of the ecosystem to the uses that are made of the data and the potential for funding. These factors accompany and promote the establishment of a DIL and the rapid deployment of its activities.

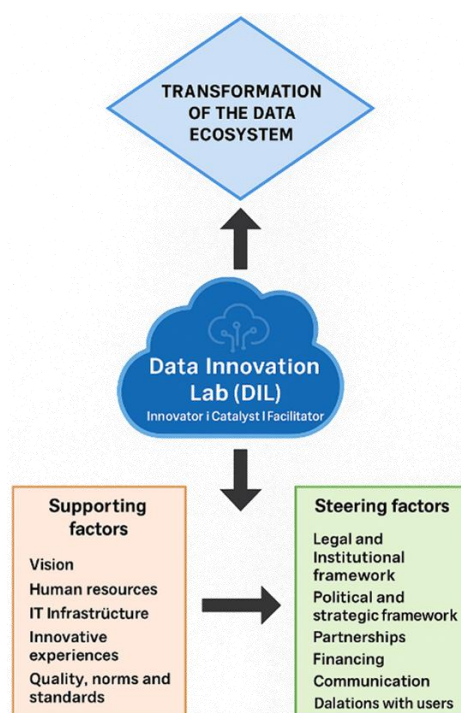


Figure 9: Possible interventions under the DIL

Proposals for interventions within the DIL framework must be part of a long-term and comprehensive approach that aims to reposition the NSO in the transformation process of the data ecosystem. This approach is based on short and long-term interventions that address the supporting and steering factors that have already been mentioned in the introduction to this report. The DIL acts as a tool in this process, which makes it more visible (within the NSO but also at the level of the data ecosystem as a whole) and gradually operationalizes it.

The ultimate objective remains the positioning of the NSO in the process of transformation of the ecosystem that has already begun, and which affects both aspects related to the supply of data (technological progress in the statistical value chain, new players and new sources of data) and those related to demand (new habits of information consumption). The installation and deployment of the DIL's activities makes it possible to link the NSO's commitment to the transformation process and to help it reposition itself in this new environment: we are moving from a context limited to the NSS to a broader context open to the entire data ecosystem. Interventions on supporting factors and guiding factors gradually feed into the deployment of the DIL in the short, medium and long term.

Carrying out the DIL diagnosis, using the maturity model, is a first step to better visualize the processes that will need to be transformed and to engage the ecosystem partners in a reflection on innovation.

7.2. Examples of activities directed at “Supporting Factors”

7.2.1. Internal vision on the transformation of the data ecosystem

- Drafting of an internal note at the NSO on the DIL vision
- Sharing and discussing the vision within NSO at the level of each department and at the level of the institution
- Innovation section/page in the NSO intranet
- Drawing a roadmap (action plan) for the implementation of the vision
- Sharing and discussing the vision with the partners in the data ecosystem

7.2.2. Human resources, awareness and training

- Data science upgrade for NSO sectoral departments
 - General awareness raising among NSO staff
 - Continuing education for some key experts in the innovation process (Master's degree or others)
 - Specialized modules as needed on certain aspects of data science
- Establishment of a competency framework for «data roles» (Chief Data Officer - CDO, Data scientist, data engineer, data architect, data stewardship, etc.).
- Encouragement of innovation (system of incentives)
- Mapping of available data science training opportunities (in-country and external)

- When possible, recruit data science specialists

7.2.3. ICT Infrastructure

- Strengthening the IT department in terms of data visualization
- Data Literacy program for other NSS actors
- Upgrading the NSO in equipment and applications
- Technology watch (in partnership with key partners)
- Development of skills related to infrastructure (computer networks, virtualization, infrastructure as code, orchestration, security, monitoring, storage technologies)
- Install the necessary infrastructure for AI applications (specialized NVIDIA graphics).

7.2.4. Innovative experiences in data management

- Finalization of ongoing innovation work on the exploitation of administrative and non- conventional data and evaluation of the results obtained
- Collection and review of innovation proposals from NSO sectors
- Collection of achievements and innovation projects of partners in the data ecosystem
- Establishment of a group of data science specialists at national level under the aegis of the NSO
- Developing a «Problem solving» competency at NSO
- Proposals for student internships (master's, PHD) to carry out test work based on innovative approaches
- Mechanisms for exchanging innovations within the data ecosystem

7.2.5. Data quality

- Formulation and implementation of a national quality framework
- Systematic monitoring of the application of quality frameworks and documentation of practices at NSO
- Technical assistance from the NSO to NSS actors for the application of norms and standards

7.3. Examples of activities directed at “Steering Factors”

7.3.1. Institutional and legal framework

- Reform of the statistical law: mobilise new non-conventional data sources and develop methods of coalition with those who collect them, review of the functioning and competences of the Statistical Authority, obligation to transfer data to the NSO for the needs of official statistics extended to all data, etc.

7.3.2. Strategic frameworks on data

- Revision of the NSDS if necessary (internalization of innovation)

- Raising awareness among NSO partners about a NDS and commitment to its preparation
- Identification of priority areas for action

7.3.3. Partnerships for innovation

- Signing of initial agreements between the NSO and a hard core of partners
- Presenting the data that are available within the ecosystem
- Testing collaboration mechanisms with the core group of partners on several themes:
 - Data availability/access
 - Data valorization
 - Exchanges and sharing of skills
 - Data science training and refresher modules
 - Technology watch
 - Exchanges and sharing of experiences
- Gradually expanding collaboration to other partners in the data ecosystem
- Specific focus on relations with universities and research (analysis)
- Specific focus on relations with municipalities (granularity)
- Establishment of regional and international partnerships

7.3.4. Communication and users' relations

- Formulation of an NSO communication policy
- Presentation sessions of the NSO to its users (skills)
- Organization of events to promote innovation and its results in statistics
- Production of notes and awareness-raising files on innovation in statistics
- Establishment of a platform for exchange on innovation in statistics
- User map
- Launching studies on the uses of data
- Development of tools for demand assessment
- Strengthening the NSO's capacity in data visualization and communication tools
- Development of visual products adapted to social media dissemination
- Improvement of the NSO website (interactivity, accessibility)
- Data awareness/training programmes for key user groups (journalists, researchers, etc.)

7.3.5. Financing innovation

- Establishment of a group of donors and specialized agencies supporting statistical innovation
- Preparation of specific projects for the financing of DIL activities to be submitted to donors

- Advocacy for increased domestic funding for statistical innovation

7.3.6. Data analytics

- Projects for the exploitation and valorization of existing data at the NSO (historical files)
- Collaborations with researchers
- Using of generative AI in data analytics
- Discussion with NSS partners for the valorization of their own data

8. Some implementation issues

8.1. A focus on the strategic vision for the DIL

In addition to clarifying the NSO's plans and ambitions for innovation, the drafting of a document on the strategic vision for the DIL has two main virtues:

- First, it allows to collect the different points of view and perspectives of the management internally (and more if the process is made participatory) and to integrate them into the process of preparing the document. This will fuel the internalization and appropriation of the vision internally, an essential condition for it to be effectively transformed into concrete actions. For innovation to take place, it is necessary to ensure the support of the staff involved in the operations, who will have the task of reflecting on why and how to innovate and, on this basis, of making and implementing proposals,
- Second, it allows the NSO's ambitions for the DIL to be clearly presented and shared, whether internally or with partners in the data ecosystem. It is therefore an essential reference document for mobilizing partnerships around the NSO project.

An indicative plan for what may be a document presenting the DIL strategic vision of the NSO is given in Appendix 4.

As mentioned above, the presentation/restitution on the DIL strategic vision is important to create a consensus on the NSO's project but also to mobilize the partnerships that are central to the concrete realization of the DIL's interventions. During the diagnostic missions in the test countries, this feedback process was discussed. It should be based on 3 complementary levels:

- Within the NSO, it is essential to strengthen internal understanding of the project to ensure that all staff are informed, engaged, and supportive. Innovation often brings changes to familiar tools and processes, which can sometimes cause uncertainty or concern. To foster a positive transition, it is crucial to provide clear information and open discussions about the implications of these changes, ensuring that everyone feels included and can find their place in the innovation process,
- With data ecosystem partners, the aim is to inform on the NSO's project and to discuss the expectations they all have with a strengthened partnership. The usual partners in the NSS for statistical work must be mobilised together with other actors in the data

ecosystem with whom working relationships are at the beginning. These new players need clear indications on the purposes of a partnership and on what it will require of them in terms of investment,

- With technical and financial partners, it is crucial to present and promote a clear framework in which they can think in advance how to contribute and reflect on an effective and complementary positioning of their support.

8.2. A focus on the issue of technology

The availability of quality IT infrastructure is certainly an asset for the development of innovation. Some applications require large computing and storage capacities that are not always available in African NSOs. However, this should not be an obstacle to advancing the NSOs' DIL agenda. There are several things to consider.

- First, the diagnoses carried out in the test countries showed that these computing and storage capacities often existed in other public structures in the countries (data centers, for example) and that, under certain conditions guaranteeing confidentiality, they could be used for official statistics,
- Then, it is always possible to work with what exists and the test countries have largely proven this. Appendix 5 of this guide presents the characteristics of what an ideal ICT infrastructure for innovation could be. There is no question of this being an essential condition for the launch of a DIL. It is rather a reference allowing NSOs to position themselves on this issue. The capacity of the IT system in the NSOs can improve gradually, depending on financial availability and opportunities, but this must be done in an articulated architecture that allows coherence and flexibility over time. It must also be combined with human skills and a secure physical environment that allow the full use of the capabilities deployed,
- Finally, it is essential to think about ICT development for innovation in a collaborative framework both for the sharing of existing capacities and for the technological watch on equipment and applications. It is also possible to consider arrangements for group purchases that would reduce acquisition costs.

8.3. Where do we stand today with the DIL initiative?

At the end of 2024, i.e. after just over two years of existence, the DIL initiative took shape on several levels:

- The implementation of DIL activities within the AfDB. Several activities have been developed since 2023 (and are still developing) that aim at building internal capacities for the understanding and the use of statistics, at developing algorithms into business processes, at producing statistics from big data and at training thematic and sector experts in data management,
- A consistency test on data from the health sector. Consistency tests are crucial for ensuring data quality in any field, but they are particularly vital in the health sector where decisions can have significant consequences. The initiative uses data from different

sources to measure development impact and track progress and to identify discrepancies or inconsistencies. The data can also be reviewed over time to ensure that trends and patterns are consistent.

- The launch of a network between French-speaking countries that have carried out the diagnostic test. This network met virtually three times since July 2024. These meetings allowed an exchange of experiences between countries but also discussion on specific themes related to the implementation of DILs. A similar network has also been established more recently among the English-speaking countries,
- The preparation of this guide,
- The holding of two «Innovate Africa» symposiums in September 2023 in Johannesburg and in October 2024 in Kigali and the participation in many workshops and conferences addressing innovation for statistics, including the meeting of the Special Technical Group of the AU on “emerging issues for statistics” (STG17).

9. Lessons learned from the test countries and experiences from other countries.

The AfDB has performed diagnostic tests in several African countries (Tunisia in November 2023, Senegal in March 2024, Côte d'Ivoire in April 2024, Nigeria in June 2024, Cameroon in September 2024, Zambia in April 2025), applying its DIL maturity model. These diagnoses were carried out by countries with the support of external expertise and were the subject of detailed individual reports⁸. They have made it possible to draw up a list of proposals and recommendations on which each country is now building its own and tailor-made DIL approach.

The following are key lessons learned from the Bank's engagement with countries where the Data-Driven Maturity Model has been applied:

9.1. A physical structure or a virtual platform to host the DIL?

African countries that have initiated the implementation of a DIL have adopted diverse approaches to establish it within their institutional frameworks, integrating previously scattered and loosely coordinated innovative initiatives. Two models exemplify the two ends of this spectrum:

1. The installation of a physical unit within a⁹ department of the NSO in charge of the DIL and of the deployment of its activities,
2. The sharing of responsibility for innovation between different NSO units via virtual

⁸ Not all the questions of the assessment questionnaire were addressed in all the countries. But overall, the questionnaire was relevant for most of the assessments carried out by the AfDB DIL initiative. The variations were very marginal and were related to the specific contexts of each country. In the countries where there was only limited progress made in using both administrative and non-conventional data, dimensions 6: Administrative data and dimension 7: Non-traditional sources have been assessed together.

⁹ Typically, this unit has been attached to the IT department or the methods department.

implementation, with only a coordination function being physically established.

Both approaches have their specific benefits and risks which are discussed in Table 2.

Table 2: Installation of the DIL

	Advantages	Risks
DIL physically installed	- Visibility of the initiative in the NSO organizational chart - Clarity of staff assignments to this unit - Simplification of financial allocation mechanisms for innovation (budget and external aid)	- Confiscation of the responsibility for innovation - Possible disengagement of technical and sectoral departments from the NSO's vision
Virtual DIL	- Commitment of the whole structure behind the NSO's vision - Investments (technical and financial) for innovation made as close as possible to operations	- Dilution of responsibilities if coordination is not ensured - Loss of overall consistency of the vision

The reality of the installation of DIL in each country will probably lie between these two extremes depending on the specific country context. Whatever the choice, it is essential to ensure that:

- The responsibility for innovation remains as close as possible to the operations within which it must be developed and internalised. In the case of the creation of a DIL unit, care must be taken to ensure that the initiative taken by the technical department to innovate in its work is not confiscated. In the case of a function-based approach, it will be essential to document and coordinate so that the results of the experiment can be analysed and disseminated within the organisation and beyond.
- All innovation investments are internally coordinated and reconciled as part of the vision. It must be possible to clearly show the results achieved as well as the benefits that have been derived from them, both internally at the NOI and with the NSO's partners in the data ecosystem.

9.2. Monitoring and evaluation

Several levels of monitoring and evaluation must be considered:

- Individual innovations.** Each experiment (or test) that will be carried out in terms of innovation by the departments of the NSO must be duly documented (monitored) and its results reviewed (evaluated) to assess its benefits and/or potential for replication. It is one of the key tasks of the DIL coordination, whether it is a physical unit or a function-based platform, to ensure that this work is carried out systematically within the NSO.

- b) The deployment of the DIL activities.** The application of the maturity model is intended to be replicated at regular intervals to monitor the overall progress made by the NSO and the entire data ecosystem in terms of innovation on data management and governance. The realization of this replica of the questioning framework can be part of a coordination within the data ecosystem that would allow collaboration on the diagnosis and discussion of the results with all the partners.

In this area, the NSO can set by itself precise objectives to be achieved (technical, financial or in terms of partnerships) in the light of which it will be able to assess in more detail the progress that has been made between two diagnoses.

9.3. Data science competency framework

NSOs in African countries have invested a lot in the past to ensure that they have quality staff trained in statistics. The technical team in the NSOs (excluding administrative and financial functions) is mainly composed of statisticians (technicians and engineers), demographers (especially when the NSO is responsible for the population census) and computer/IT scientists (at all levels). These backgrounds are crucial in traditional statistical production, but today they do not cover all the needs generated by the transformation of the nature of the work in data ecosystems. There are, in general, very few data science specialists within NSOs and other public structures working on data. There are several reasons for this:

- First, it is a new concern, and these data scientists are still rare on the job market,
- Secondly, there is strong competition for the employment of these experts with the private sector, which can provide them with better salaries and better prospects.

The situation is changing rapidly and there are more training structures that now offer specific curricula in data science. Despite this, their recruitment into the civil service is hampered by the absence of a competency framework in this area that would give them a clear status in the salary scales and career profiles in the civil service.

9.4. Mobilisation of innovative projects

The diagnostic missions in the test countries all showed that the technical departments of the NSOs had fairly clear and precise ideas on how they could improve their practices and tools, developing and using innovation. To make these ideas a reality, several elements were missing:

- A more in-depth knowledge of the technical potential offered by data science, whether in terms of IT applications or organisational modalities. There is therefore a demand for information, awareness and training on these aspects to fully unleash the existing innovative potential in the technical departments,
- Rapid and targeted access to specialized expertise to compensate for technical and capacity gaps may certainly help. In terms of human capacities, very positive approaches were explored, such as the mobilization of students/trainees or collaboration with experts from the academic and research worlds. In terms of

equipment, collaboration with the NSO IT department is not always simple and organized,

- Channels organized to present innovation projects from technical departments, evaluate them and finance their implementation are not clearly set and developed. Most of the time, it is individual initiative and goodwill that drive projects forward,
- Availability in the technical departments to (temporarily) turn away from their daily tasks and devote themselves to innovation is scarce. Since there is «training time» that is increasingly allocated to the staff of organizations, innovation time» should be considered.

It would be possible to envisage this set in a specific function of the DIL organised around the mobilisation of innovative projects within the NSO, their evaluation and, in the case of promising projects, their implementation and monitoring. An internal (light) committee could be set up that would regularly review the proposed projects, and the results obtained.

Eventually, the process put in place should be extended to NSO partners within the country's data ecosystem.

9.5. Partnerships and coalitions

All the tests carried out in the framework of the AfDB-DIL initiative have shown the importance of partnership and of the mobilization of coalitions for innovation within the data ecosystem. The transformation of the ecosystem will be profound and will quickly have an impact on the survival of official statistics. In most countries, the NSO is not alone in preserving quality gains and setting the entire data ecosystem on the path to transformation. It is essential to bring on board other structures that will advance the common agenda.

In data ecosystems, NSO partners do not all have the same levels of development or the same agendas. It would be difficult to try to bring quickly such different structures into a coordinated approach. It is therefore advisable to move forward in stages and start the coalition process with a limited group (hard core) that shares the same concerns about innovation. With these privileged partners, it will be necessary to build tangible results that will be as many arguments as possible to be used at a later stage to mobilize other structures.

The AfDB-DIL diagnostic missions found that in each country, there are core partners ready to be mobilised behind the NSO's DIL agenda.

10. Some good practices from test countries

There are many examples of innovation in the countries who participated in the AfDB-DIL test and they concern all stages of the value chain of statistics. There are many ongoing projects that are avenues for integrating more innovation into current operations. Below are some examples.

System of «Use cases» - example from Senegal. Many «use cases» are underway. For this work, the ANSD relies on interns from its statistical school (ENSAE - about ten so far).

The most elaborate project concerns the use of mobile phone data on remittances to support predictions on key social and poverty indicators. Other use cases have also been launched in cooperation with sectoral partners within the NSS. Also, worth mentioning is work with the Ministry of the Economy and Finance to change the methods used for household targeting (substitution of surveys and censuses by algorithmic methods). The idea is to now test the reliability of the results obtained with these models compared to traditional tools by launching a light survey. A review of the results obtained will be made to determine the extent to which it is possible to adjust current practices.

Mobile phone data – examples from Tunisia, Côte d'Ivoire and Senegal. The INS of Tunisia is working on the use of mobile phone data for the calculation of indicators on internal migration. Tunisia is quite advanced in its relations with local operators and the consultation process launched by the INS with them is very exemplary (high-level meetings). The INS is supported in this project by the World Bank. The project has also aroused the interest of other countries (Senegal in particular, which is also working on the use of these data sources). It should be noted that the INS intends to use the current census to test the robustness of the estimates made with mobile phone data.

In Côte d'Ivoire, AnStat has developed exchanges with operators on the use of mobile data for statistics in parallel with the overhaul of its employment surveys. A joint discussion has been launched with the company “Orange” on the modalities of cooperation for statistics with a specific focus on the cost-sharing aspects linked to the use of these sources.

In Senegal, the ANSD has initiated discussions with SONATEL, the local telephone company, to sign a technical cooperation agreement at the beginning of 2025.

Mobilization of NSO partners – example from Côte d'Ivoire. The conceptualization of an «innovation café», which would take the name «Let's talk stats», should make it possible to exchange widely in the country on innovation and the benefits to be derived from it in terms of data management and data governance. The first meeting should be organized in the first quarter of 2025 and mobilize widely beyond ANSTAT. The project is supported by the World Bank.

Data Strategy – example from Nigeria. Multiple national strategies are available or under development that directly concern statistical development (NSDS III 2024-2028 currently being finalized) or key areas of support (innovation, digitalization, data security, ...) for the installation and deployment of the DIL. Nigeria is one of the few African countries that has developed a national data strategy¹⁰. The benefits to be derived from these strategies are manifold and can support and accelerate the deployment of innovation across the official statistics value chain. Table 3 is an indicative and non- exhaustive list of policy and legal texts that have been developed by Nigeria in these key areas for the transformation of data ecosystems in recent years.

¹⁰ It should be noted that Senegal has also developed a data strategy. It was the subject of a kick-off seminar in March 2023.
Summarized version: chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://datapopalliance.org/wp-content/uploads/2018/01/Resume-de-Strategie-Nationale-des-Donnees-du-Se_230724_172910.pdf **Seminar:** chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://www.numerique.gouv.sn/sites/default/files/Workshop%20Strat%C3%A9gie%20des%20Donn%C3%A9es%2023_24%20Mars%20Document%20Presentation%20DPA.PDF.

Table 2: Strategy documents in Nigeria

National data strategy 2022	chrome-extension://efaidnbmnnnibpcajpcgglefindmkaj/ https://nitda.gov.ng/wp-content/uploads/2022/11/Final-Draft-National-Data-Strategy.pdf
National Data Protection Act 2023	chrome-extension://efaidnbmnnnibpcajpcgglefindmkaj/ https://placng.org/i/wp-content/uploads/2023/06/Nigeria-Data-Protection-Act-2023.pdf
National Digital Economy Policy and Strategy 2020 – 2030	chrome-extension://efaidnbmnnnibpcajpcgglefindmkaj/ https://ncc.gov.ng/accessible/documents/883-national-digital-economy-policy-and-strategy/file
Nigerian National Broadband Plan (NNBP) 2020 - 2025	https://ngfrepository.org.ng:8443/handle/123456789/3349
National Cloud Computing Policy	chrome-extension://efaidnbmnnnibpcajpcgglefindmkaj/ https://nitda.gov.ng/wp-content/uploads/2020/11/NCC-Policy_New1.pdf
Nigeria Digital Economy Ecosystem Development Strategy (N-DEEDS)	chrome-extension://efaidnbmnnnibpcajpcgglefindmkaj/ https://nitda.gov.ng/wp-content/uploads/2022/11/Final-Draft-National-Data-Strategy.pdf
Nigeria e-Government Interoperability Framework (Ne-GIF)	chrome-extension://efaidnbmnnnibpcajpcgglefindmkaj/ https://nitda.gov.ng/wp-content/uploads/2020/11/Ne-GIFFinal1.pdf
Nigeria Open Data Policy (NODP)	chrome-extension://efaidnbmnnnibpcajpcgglefindmkaj/ https://neiti.gov.ng/cms/wp-content/uploads/2021/08/NEITI-Open-Data-Policy.pdf

Use of satellite data – examples from Cameroon, Côte d'Ivoire and Tunisia. All three countries have initiated tests to use satellite data for statistical purposes. This is a very promising field, but it faces difficulties related to the cost of mobilizing quality images.

The ANSTAT of Côte d'Ivoire works with images taken from Google Earth¹¹ and Sentinel 2A¹² (Copernicus) which are available free of charge. The level of performance is interesting, but the resolution is still quite low and does not allow for precision work. On an experimental basis, contacts have been made with the company “Airbus”, which has more precise images but whose cost is high. The availability of this new source and its potential was tested with an initial (limited) financial allocation. A report has been produced to make recommendations to the DG in this area.

¹¹

<https://earth.google.com/web/@49.66263298,6.10828197,307.24250193a,82658.29410556d,35y,0h,0t,0r/data=CgRCAggBOgMKATBCAggAS-g0IARAA>

¹² https://browser.dataspace.copernicus.eu/?zoom=12&lat=45.34382&lng=12.3888&themeId=DEFAULT-THEME&visualizationUrl=U2FsdGVkX18%2Bm70dbb%2BNGGbzKz9X6Orl9rKTTGnJb9yG%2FT%2BkzWqTKumdPQIE5StoBka56Nn%2Bz%2F55thCKz09RiR0FOGfVt-dcRVKT6DZiluWinoFkTv%2FJF83plF4jTcrU&datasetId=S2_L1C_CDAS&fromTime=2023-02-07T00%3A00%3A00.000Z&toTime=2023-02-07T23%3A59%3A59.999Z&layerId=1_TRUE_COLOR&demSource3D=%22MAPZEN%22&cloudCoverage=10&dateMode=SINGLE

The INS of Tunisia worked for the general population and housing census in cooperation with the National Centre for Remote Sensing (NTC) which provided the satellite data. A cooperation MoU has been signed, and the INS will now use the census outputs to assist the NTC in setting up a national register of addresses. On the other hand, as part of the UN D4N and in cooperation with the Ministry of Environment and Agriculture, the INS has benefited from capacity building and access to satellite data to calculate some SDG indicators.

The INS of Cameroon organized in September 2024 a workshop on the use of satellite data for population statistics and the mobilization of primary data on education by the INS. It was attended by several members of the NSS and other actors of the data ecosystem, including universities and private companies. This was an opportunity to exchange views and experiences.

On this subject, the AfDB's DIL initiative could organize a specific exchange of experience between African countries in the course of 2025.

Web scraping - examples from Ivory Coast, Cameroon and Senegal. In these three countries, the data available on the Internet have been used mainly to supplement the surveys which are made through price surveys. In Côte d'Ivoire, the ANSTAT also tests these data to supplement employment statistics (data on job vacancies).

11. Can a simplified version of the model be adopted?

The maturity model that has been developed by the AfDB covers a wide field of investigation that needs to be analyzed in detail to properly understand not only the existing potential to ensure an effective transformation of the data ecosystem through innovation, but also the points of fragility that could slow down the deployment of DIL activities. It nevertheless remains perfectly manageable, as proven by the diagnoses carried out in the test countries, with the support of external expertise - the completion of the questionnaire, the discussion on the state of play and then the identification of a business plan will have taken, on average, a little more than two/three months per country (time between the sending of the questionnaire to the country and the submission of the diagnostic draft report). The external expertise certainly made it possible to mobilize the teams in the NSOs and their partners more quickly than if the process had taken place internally. The expert's mission, which never lasted more than 5-6 days, was an opportunity to mobilise all the teams within and outside the NSOs.

However, it is possible to launch the diagnostic process based on a simplified model which is presented in **Appendix 6**. The themes are identical, but it is proposed to answer each of them globally without going into the details of the complete maturity model. For each theme, the reference elements that should make it possible to assess the situation are given. It is even recommended to only focus on major areas of intervention on each theme according to the country's specific position on the issue.

The AfDB DIL team strongly encourages the countries to adopt the full questionnaire for a comprehensive initial assessment of the degree of preparedness of their statistical system

for innovation. The diagnostic may then be realised with the simplified model as a pilot and monitoring tool to assess regularly the progress made towards the transformation of the data ecosystem.

12. What is the future for the AfDB's DIL initiative?

The overall objective of the AfDB is to support the deployment of the DIL agendas in Africa (starting with the countries participating in the AfDB-DIL initiative and gradually expanding to others that will want to join) through a regional dialogue between countries on the transformation of their data ecosystems and the role that innovation can play in this process. This objective is part of a desire to broaden the scope of collaboration and exchange between NSOs on key issues affecting their evolution such as accessibility of data, the valorization of existing statistics, the use of new data sources for statistics and the dissemination of technological progress in data ecosystems at all levels of the value chain.

A future for the initiative could be based on the following main elements:

1. Direct support for the implementation of DIL action plans in African countries and the mobilization of other technical and financial support,
2. Expanding the number of country diagnoses, using various approaches (with or without external expertise). This would make it possible to multiply the experiences on which to learn lessons,
3. Supporting follow-up diagnostics (mainly self-diagnostics) in the countries having already realised the initial assessment as to measure and monitor progress made,
4. Strengthening networking among countries that have developed a DIL initiative, with the AfDB or otherwise. The experience of regular meetings among the test countries has shown that there is great interest in the exchange of practices in this constantly evolving field,
5. The mobilization of continental actors behind the conclusions and recommendations of the «Innovate Africa» Symposium and the formulation of regional strategic frameworks in areas related to data science.

ANNEXES

Annex 1: The AfDB-DIL Maturity Model Questionnaire

Annex 2: Synthetic presentation of the results of the DIL diagnostic

Annex 3: Presentation of the recommendations emerging from the diagnostic

Annex 4: A format for the formulation of the NSO DIL strategic vision

Annex 5: ICT reference framework

Annex 6: A simplified model

Annex 7: Comprehensive Overview of AfDB Data Innovation Lab Activities (2022–2025)

Annex 1: The AfDB-DIL maturity model questionnaire

Assessing the level of preparedness of the country to transform the data ecosystem and to establish a Data Innovation Lab (DIL): preparing the diagnostic mission.

The diagnostic mission aims at assessing the level of preparedness of the national statistical system to embark in a process of transformation based on innovation, the first step of the process being the setting-up of a Data Innovation lab hosted by the national statistical Institute. The establishment of the DIL may be virtual (a platform of exchange and information) or physical (a taskforce or a Unit).

The diagnostic will lead to concrete and operational proposals that may feed the deployment of immediate activities by the DIL or the development of activities with a longer-term perspective towards the transformation of the data eco-system system. The activities, short and long term, should be adapted to the country context.

Below are the main issues and questions that will be discussed during the diagnostic mission. Indications are also given regarding the basic documentation to assemble and make available to the mission. These questions and the list of documents are not exhaustive and any suggestions by the statistical office for other issues to be dealt with is very welcome.

The questions are presented along 12 thematic issues.

- 1. The leadership of the National Statistical Office (NSO) is convinced that a transformation of the data eco-system is necessary, and the management and staff mostly share and support this vision.**

We want to assess to what extent the management and the staff of the NSO understand why a Data Innovation Lab is established and for what reason.

To feed the assessment:

- Declarations from the DG and of the management of the NSO during internal meetings, in internal documents, at the occasion of the Statistical Council (or similar body) or at the occasion of events at national, regional and international levels,
- Interviews with the NSO's management and other staff members dealing with innovation and data science.

N°	Questions	Answers
1.1.1	What are, within the NSO, the practices with the dissemination of innovation for statistics and with exchanges on the transformative vision of the data eco-system?	
1.1.2	What role does the middle management play in this transformative vision?	
1.1.3	What is the level of support that the staff gives to the transformative agenda?	
1.1.4	What the main resistances to change within the NSO?	
1.1.5	Which level of initiative is given to the staff of the NSO for the development and implementation of innovation and changes in practices based on data science?	

N°	Questions	Answers
1.2.1	How is shared the transformative vision by the management/leadership of the NSO (not only the DG?	
1.2.2	How is the transformative vision made available and accessible to the NSO's staff (documents, meetings, ...)?	
1.2.3	Are there internal documents presenting the NSO transformative vision?	
1.2.4	Are there "champions" within the NSO that could relay, disseminate, and explain the transformation to others?	

2. The NSO has prepared a strategic vision for the transformation of the data eco-system which is being implemented progressively. This vision is supported by the Government.

We want to assess to what extent the NSO but also the Government understand the challenges for the data eco-system posed by the rapid development of digitalisation and the fast increase of data availability (Open data)

To feed the assessment:

- Strategic documents from the NSO (NSDS and others) and/or from the Government (National Data strategy or others),
- Operational plan at short and long terms,
- Interviews with NSO management and other staff. Contact with other institutional partners including the Ministry responsible for statistics.

N°	Questions	Answers
2.1.1	Does the NSDS clearly mention the need for the transformation of the data Eco-system and in particular the use of non-statistical sources of data (administrative data, big data ...)?	
2.1.2	is there a corpus of experiences regarding the use of these sources in the NSO and in the country?	
2.2	Are statistics regularly used or cited as a reference in Government strategic documents (Statistical section in general or sector strategic documents, contributions of the NSO to the writing of these documents)?	
2.3	Is there a recent national data strategy?	
2.4.1	How is the organisation of the NSO: hierarchical, compartmentalised, cross-sectional with task forces ...?	-
2.4.2	How is organised the coordination among the services within the NSO?	
2.4.3	How does this cooperation work in particular for some transversal issues that are thematic (such as gender) or technical (data processing, dissemination ...)?	
2.4.4	Which autonomy does the NSO have to modify its internal organization and structure?	

3. A legal and institutional framework that is clearly giving to the NSO responsibilities in terms of coordination of the national statistical system and of all questions related to the use of « open data » for statistics.

We seek information of the room of manoeuvre that the NSO has for making its responsibility in terms of coordination evolve beyond the boundary of the National statistical system and cover the whole data eco-system.

To feed the assessment:

- Statistical law,
- Legal framework related to the Statistical Council/Committee if any,
- Legal framework related to the NSO,
- Documents administrative and political organisation of the country,
- Interviews NSO management. Contact with the Statistical Council/Committee and with some structures of the NSS having relation with the NSO (interconnectivity between data banks, secured data exchange, partnership agreement ...).

N°	Questions	Answers
3.1.1	Is the statistical law clear on the central coordination role of the NSO?	
3.1.2	How does it work in practice?	
3.1.3	Is NSO able to get the data it asks from its partners in the NSS (all data, all partners)?	
3.1.4	Is NSO allowed to take direct contacts with actors from the private sector to collect data beyond regular survey work?	
3.1.5	How is organised data exchange between the NSO and its partners in the NSS?	
3.2.1	How is organised the exchange of experience within the NSO and within the NSS?	
3.2.2	Did these exchanges impact practices in data management within the NSO and within the NSS?	
3.3.1	How does the NSO organise/provide « data stewardship » internally, within the NSS and beyond (data are accessible, usable, safe, and trusted)?	
3.3.2	Is there a legal framework to organise data stewardship in the country?	
3.4	What degree of connectivity has been reached within the data eco-system between the various actors and the tools for data collection, processing, storage, validation, analysis and dissemination?	
3.5.1	What is the role of the Statistical Council/Committee? (Is it active in the promotion of statistics in the country?)	
3.5.2	Does the Committee have a concrete involvement in the dissemination of norms and in the coordination of the NSS?	
3.5.3	Does the Council/Committee advise the Government for the validation of the statistical strategy?	
3.6.1	What role does the NSO play in the monitoring of the SDGs?	
3.6.2	Does the NSO have a specific responsibility (legal or technical) for the production, coordination, and dissemination of the SDGs indicators?	

4. Staff trained and/or responsive to data science within the NSO

We seek to identify human resources as an effective asset for the implementation of the transformative agenda.

To feed the assessment:

- Documents on human resources at the NSO (quantity et quality) and on the available capacities in IT and data science
- Training programmes at NSO (data science)
- List of national (and regional) organisation delivering training on data science (relevance of their curriculum)
- Interviews with HR and training unit at NSO. Contact with national training institutions delivering courses on data science.

N°	Questions	Answers
4.1	What are the capacities of the NSO in Data science and Connex thematic (number of specialists and proportion related to total staff)?	
4.2.1	What are the internal processes used to strengthen staff capacities in data science?	
4.2.2	Do NSO training plans systematically include data science?	
4.2.3	Is there a special mechanism within NSO to exchange experiences in data science?	
4.3.1	Did NSO established partnerships with universities and training centres at national level in data science?	
4.3.2	What kind of partnerships: involvement of NSO staff as trainers, hosting of scholars at the NSO ...?	
4.3.3	Same questions as 4.3.1 but concerning partnerships and exchange of good practices with other Statistical institutes, regional and/or international organisations?	

5. Powerful and modern IT equipment and applications in the NSO

We want to know if the IT equipment available at NSO is sufficient to initiate and strengthen the transformative agenda.

To feed the assessment:

- Review of the available IT equipment at NSO,
- Review of the potentialities offered on the NSO website and of the presence of NSO on the social medias,
- Interviews with staff involved in IT and the development of applications. Contact with staff working on the website and on the communication on the social medias.

N°	Questions	Answers
5.1.1	How recent and modern is the IT infrastructure at NSO?	
5.1.2	Same question regarding the applications for data processing, data exchange and dissemination?	
5.1.3	Same question regarding the website and the use of social medias?	
5.1.4	How is security dealt with at NSO?	
5.2.1	What are the tools that are used by NSO to carry out its surveys and censuses?	
5.2.2	Does NSO use complex software and applications for data analysis?	
5.2.3	Does NSO use complex tools for the dissemination of data?	
5.3	What is the level of sophistication of the tools available on the website to consult and analyse the data (tables, graphs, data extraction ...)?	
5.4.1	What is the level reached by NSO regarding the use of administrative data for statistics (number of active partners, regularity, and security of data exchange, ...)?	
5.4.2	Same question regarding the use of registers?	
5.4.3	What role does NSO play in supporting partners in the NSS for data management?	

6. Experience at NSO related to the use of administrative data for statistics

We seek information on the investment made by the NSO on the mobilisation and use of administrative data and on the good practices developed in this area.

To feed the assessment:

- Internal NSO documents (partnerships agreements ...),
- Review of the modalities for the processing and the use of registers and administrative data (population, businesses, other themes ...),
- Interviews with NSO staff and with key partners in this area (ministries and other public agencies collaborating with NSO).

N°	Questions	Answers
6.1	Please describe the main experiences initiated in this area by NSO (which partner, depth of the partnership ...)?	
6.2	Any problem faced in the partnerships, if any (secured data transfer, classifications, norms and methodologies, connectivity, calendars ...)?	

7. Experience at NSO related to the use of non-conventional data sources

We seek information on the investment made by the NSO on the mobilisation and use of non-conventional sources of data for statistics and on the good practices developed in this area.

To feed the assessment:

- Mapping of the available non-conventional sources of data in the country, if it exists,
- Internal NSO documents (partnerships agreements ...),
- Interviews with NSO staff and with key partners in this area.

N°	Questions	Answers
7.1	Are there experiences that have been carried out in the area of open data: satellite imagery, data from drones and sensors, people generated data ...?	
7.2	Same question regarding specific areas such as mobile data and transport data?	

8. Organised partnerships with data owners in the public and private sectors on data science

We seek information on the contacts taken and partnerships developed by NSO with data owners from the public and private sectors in the area of data science.

To feed the assessment:

- Internal NSO documents (partnerships agreements ...),
- Interview with NSO management and administrative services.

N°	Questions	Answers
8.1.1	What role does play NSO in disseminating data management practices within the NSS?	
8.1.2	Are there partners that are more open to data science and its potential for data management?	
8.1.3	What specific collaborations has been developed regarding training and data exchange?	
8.2	Are there in the country platforms of data exchange and integration other than the one related to the SDG monitoring?	
8.3.1	What is the room of maneuverer for NSO to establish contacts with partners from the private sector and the civil society?	
8.3.2	Are special legal authorisations necessary to collect and exchange data with them?	

9. Special considerations brought to quality and to international norms and methods.

We seek information on the level of compliance of NSO practices with international/regional norms and standards. We also want to know how this compliance is shared by NSO partners in the NSS. We are also interested in quality management within the NSO operations and how this consideration is shared by partners in the NSS. Strict quality standards may contribute to the reputation of the statistical system; it can also contribute to facilitating the development of the transformative agenda.

To feed the assessment:

- « Repository » of information on methods (GDDS, SDDS),
- Internal NSO documents on quality and compliance with international norms,
- Interviews with NSO staff involved in these areas.

N°	Questions	Answers
9.1.1	What quality models are currently used in NSO (UNQAF, GSBPM ...)?	
9.1.2	What is the degree of application of these models in NSO operations?	

9.1.3	Is the level of application monitored? If yes, how does that work?	
9.2.1	Is there a culture of excellence at the NSO that is promoted by the management and supported by the staff?	
9.2.2	Does NSO play a role in disseminated quality concerns and methods within the NSS?	
9.3.1	Is NSO involved in a regional/international scheme for exchanging experiences and good practices on quality?	
9.3.2	If yes, please briefly describe the content and the state of these experiences?	
9.4.1	What is the level of compliance of NSO with international recommendations on norms and standards?	
9.4.2	How is this level of compliance shared with partners in the NSS?	
9.5	Does NSO provide support to its partners in the NSS to explain the norms and to assist in their application?	
9.6.1	What reputation does NSO have in this area internationally?	
9.6.2	What is its implication in international and regional discussions and work groups in this area?	
9.6.2	To what extent does he/she participate in technical debates/discussions at these levels?	

10. A strong concern given to meeting the needs of the users.

We are looking for evidence regarding the effectiveness of the relations developed with the users and the impact this can have on their confidence and trust in official statistics.

To feed the assessment:

- Posts in social medias and return on satisfaction from surveys,
- NSO internal documents (diffusion, communication ...),
- Interviews with staff in charge of dissemination and communication at NSO. Contacts with journalists and representatives from groups of users.

N°	Questions	Answers
10.1.1	Does NSO have a communication policy that concerns the production and dissemination of data, but also other issues linked to NSO work (professionalism, objectivity, social responsibility ...)?	
10.1.2	How is managed « branding » at NSO?	
10.2.1	What are the main relations established between NSO and its users (satisfaction surveys, users' groups, feed-back posts on the website ...)	
10.2.2	How are returns from the users considered in NSO operations (examples)?	
10.3.	Are there examples to assess the reactivity and agility of NSO to adjust to a change in the context and to translate this into its current operations (example: What did NSO do to answer to the information needs generated by COVID)?	
10.4.1	What is the availability of data in the country and what are the priority issues for NSO and the NSS in terms of data collection and production?	
10.4.2	Are there important domains and issues that are badly or not covered?	
10.4.3	What is NSO policy for the access to its data?	

10.5.1	Is there a specific demand in the country for data that are multi-dimensional and/or for data that cover several themes (poverty, vulnerability, climate change ...)?	
10.5.2	How does NSO answer to these demands?	
10.6	Do you know the extent of the use of NSO data for decision making by the public administration and other users (private sector, civil society, donor community ...)?	

11. A strong and acknowledge capacity in research and complex analysis at the NSO and/or in relation with research centers and academics

We want to know what level the NSO reached in complex data analysis. We also want to know about the availability and use of data for analysis.

To feed the assessment:

- Internal NSO documents on analyses and complex studies realised,
- Interview with NSO staff involved in research alone or in collaboration with universities and/or research centres. Contacts with some of these external partners in university.

N°	Questions	Answers
11.1	How often does NSO work with universities and research centres on socio-economic and demographic analysis?	
11.2	Did NSO develop a strong capacity in the design of applications that rely on new technologies such as AI or machine learning (give example if any)?	
11.3.1	Did NSO work, alone or with partners, in data study going beyond contextual analysis? If yes, with which partners?	
11.3.2	Which research tools are used by NSO (prediction, forecast/nowcast and prospective)?	
11.4.1	What are the modalities for researchers to be granted access to microdata available at NSO?	
11.4.2	How is confidentiality of individual data protected by the NSO?	
11.5	What is the national potential for socio-economic research and studies (number of organisations involved in research, number of researchers working on NSO/NSS data)?	

12.Active technical and financial support that may potentially be mobilised to accompany the implementation of the transformative agenda of the data eco-system

We want to assess the existing potential support which could be made available along the implementation of the transformative agenda and in particular for the setting up of the DIL.

To feed the assessment:

- Internal documents regarding the financial assistance received for statistics in the last few years from external donors and the technical support from international and regional partners.
- Interview with NSO staff. Contact with the ministry in charge of external support and with donors/organisations supporting innovation in the country.

N°	Questions	Answers
12.1	Is there a group of supporting donors for statistics?	
12.2	What are the main projects and programmes that will be funded by external support in the next few years and through which some activities of the DIL could be covered?	
12.3	What is the support that the NSO can expect to get from the national budget for its statistical programme in the next three years?	

Annex 2: Format for the synthetic presentation of the results of the DIL Diagnostic

(The last column records the color code that has been determined through the detailed analysis of the questionnaire: green, orange and red. Grey means that there is no answer given at the time the analysis is made or that the question is not relevant in the specific country context.)

1. Vision by the management on the transformation of the data eco-system		
1.1.1	What are, within the NSO, the practices with the dissemination of innovation for statistics and with exchanges on the transformative vision of the data eco-system?	
1.1.2	What role does the middle management play in this transformative vision?	
1.1.3	What is the level of support that the staff gives to the transformative agenda?	
1.1.4	What are the main resistances to change within the NSO?	
1.1.5	Which level of initiative is given to the staff of the NSO for the development and implementation of innovation and changes in practices based on data science?	
1.2.1	How is shared the transformative vision by the management/leadership of the NSO (not only the DG?	
1.2.2	How is the transformative vision made available and accessible to the NSO's staff (documents, meetings, ...)?	
1.2.3	Are there internal documents presenting the NSO transformative vision?	
1.2.4	Are there "champions" within the NSO that could relay, disseminate, and explain the transformation to others?	
2. Strategic vision on the transformation of the data ecosystem		
2.1.1	Does the NSDS clearly mention the need for the transformation of the data Eco-system and in particular the use of non-statistical sources of data (administrative data, big data ...)?	
2.1.2	is there a corpus of experiences regarding the use of these sources in the NSO and in the country?	
2.2	Are statistics regularly used or cited as a reference in Government strategic documents (Statistical section in general or sector strategic documents, contributions of the NSO to the writing of these documents)?	
2.3	Is there a recent national data strategy?	
2.4.1	How is the organisation of the NSO: hierarchical, compartmentalised, cross-sectional with task forces ...?	
2.4.2	How is organised the coordination among the services within the NSO?	
2.4.3	How does this cooperation work in particular for some transversal issues that are thematic (such as gender) or technical (data processing, dissemination ...)?	
2.4.4	Which autonomy does the NSO have to modify its internal organization and structure?	

3. Legal and Institutional framework		
3.1.1	Is the statistical law clear on the central coordination role of the NSO?	
3.1.2	How does it work in practice?	
3.1.3	Is NSO able to get the data it asks from its partners in the NSS (all data, all partners)?	
3.1.4	Is NSO allowed to take direct contacts with actors from the private sector to collect data beyond regular survey work?	
3.1.5	How is organised data exchange between the NSO and its partners in the NSS?	
3.2.1	How is organised the exchange of experience within the NSO and within the NSS?	
3.2.2	Did these exchanges impact practices in data management within the NSO and within the NSS?	
3.3.1	How does the NSO organise/provide « data stewardship » internally, within the NSS and beyond (data are accessible, usable, safe, and trusted)?	
3.3.2	Is there a legal framework to organise data stewardship in the country?	
3.4	What degree of connectivity has been reached within the data eco-system between the various actors and the tools for data collection, processing, storage, validation, analysis and dissemination?	
3.5.1	What is the role of the Statistical Council/Committee? (Is it active in the promotion of statistics in the country?)	
3.5.2	Does the Committee have a concrete involvement in the dissemination of norms and in the coordination of the NSS?	
3.5.3	Does the Council/Committee advise the Government for the validation of the statistical strategy?	
3.6.1	What role does the NSO play in the monitoring of the SDGs?	
3.6.2	Does the NSO have a specific responsibility (legal or technical) for the production, coordination, and dissemination of the SDGs indicators?	
4. Capacities: Human resources		
4.1	What are the capacities of the NSO in Data science and Connex thematic (number of specialists and proportion related to total staff)?	
4.2.1	What are the internal processes used to strengthen staff capacities in data science?	
4.2.2	Do NSO training plans systematically include data science?	
4.2.3	Is there a special mechanism within NSO to exchange experiences in data science?	
4.3.1	Did NSO established partnerships with universities and training centres at national level in data science?	
4.3.2	What kind of partnerships: involvement of NSO staff as trainers, hosting of scholars at the NSO ...?	
4.3.3	Same questions as 4.3.1 but concerning partnerships and exchange of good practices with other Statistical institutes, regional and/or international organisations?	

5. Capacities: IT Infrastructure and application		
5.1.1	How recent and modern is the IT infrastructure at NSO?	
5.1.2	Same question regarding the applications for data processing, data exchange and dis- semination?	
5.1.3	Same question regarding the website and the use of social medias?	
5.1.4	How is security dealt with at NSO?	
5.2.1	What are the tools that are used by NSO to carry out its surveys and censuses?	
5.2.2	Does NSO use complex software and applications for data analysis?	
5.2.3	Does NSO use complex tools for the dissemination of data?	
5.3	What is the level of sophistication of the tools available on the website to consult and analyse the data (tables, graphs, data extraction ...)?	
5.4.1	What is the level reached by NSO regarding the use of administrative data for statistics (number of active partners, regularity, and security of data exchange, ...)?	
5.4.2	Same question regarding the use of registers?	
5.4.3	What role does NSO play in supporting partners in the NSS for data management?	
6. Use of administrative data for statistics		
6.1	Please describe the main experiences initiated in this area by NSO (which partner, depth of the partnership ...)?	
6.2	Any problem faced in the partnerships, if any (secured data transfer, classifications, norms and methodologies, connectivity, calendars ...)?	
7. Use of non-conventional data sources for statistics		
7.1	Are there experiences that have been carried out in the area of open data: satellite imagery, data from drones and sensors, people generated data ...?	
7.2	Same question regarding specific areas such as mobile data and transport data?	
8. Organised and efficient partnerships		
8.1.1	What role does play NSO in disseminating data management practices within the NSS?	
8.1.2	Are there partners that are more open to data science and its potential for data management?	
8.1.3	What specific collaborations has been developed regarding training and data ex- change?	
8.2	Are there in the country platforms of data exchange and integration other than the one related to the SDG monitoring?	
8.3.1	What is the room of manoeuvre for NSO to establish contacts with partners from the private sector and the civil society?	
8.3.2	Are special legal authorisations necessary to collect and exchange data with them?	

9. Norms and standards, quality		
9.1.1	What quality models are currently used in NSO (UNQAF, GSBPM ...)?	
9.1.2	What is the degree of application of these models in NSO operations?	
9.1.3	Is the level of application monitored? If yes, how does that work?	
9.2.1	Is there a culture of excellence at the NSO that is promoted by the management and supported by the staff?	
9.2.2	Does NSO play a role in disseminated quality concerns and methods within the NSS?	
9.3.1	Is NSO involved in a regional/international scheme for exchanging experiences and good practices on quality?	
9.3.2	If yes, please briefly describe the content and the state of these experiences?	
9.4.1	What is the level of compliance of NSO with international recommendations on norms and standards?	
9.4.2	How is this level of compliance shared with partners in the NSS?	
9.5	Does NSO provide support to its partners in the NSS to explain the norms and to assist in their application?	
9.6.1	What reputation does NSO have in this area internationally?	
9.6.2	What is its implication in international and regional discussions and work groups in this area?	
10. Users' relations and communication		
10.1.1	Does NSO have a communication policy that concerns the production and dissemination of data but also other issues linked to NSO work (professionalism, objectivity, social responsibility ...)?	
10.1.2	How is managed « branding » at NSO?	
10.2.1	What are the main relations established between NSO and its users (satisfaction surveys, users groups, feed-back posts on the website ...)	
10.2.2	How are returns from the users considered in NSO operations (examples)?	
10.3.	Are there examples to assess the reactivity and agility of NSO to adjust to a change in the context and to translate this into its current operations (example: What did NSO do to answer to the information needs generated by COVID)?	
10.4.1	What is the availability of data in the country and what are the priority issues for NSO and the NSS in terms of data collection and production?	
10.4.2	Are there important domains and issues that are badly or not covered?	
10.4.3	What is NSO policy for the access to its data?	
10.5.1	Is there a specific demand in the country for data that are multi-dimensional and/or for data that cover several themes (poverty, vulnerability, climate change ...)?	
10.5.2	How does NSO answer to these demands?	
10.6	Do you know the extent of the use of NSO data for decision making by the public administration and other users (private sector, civil society, donor community ...)?	

11. Research and data analytics		
11.1	How often does NSO work with universities and research centres on socio-economic and demographic analysis?	
11.2	Did NSO develop a strong capacity in the design of applications that rely on new technologies such as AI or machine learning (give example if any)?	
11.3.1	Did NSO work, alone or with partners, in data study going beyond contextual analysis? If yes, with which partners?	
11.3.2	Which research tools are used by NSO (prediction, forecast/nowcast and prospective)?	
11.4.1	What are the modalities for researchers to be granted access to microdata available at NSO?	
11.4.2	How is confidentiality of individual data protected by the NSO?	
11.5	What is the national potential for socio-economic research and studies (number of organisations involved in research, number of researchers working on NSO/NSS data)?	
12. Technical and financial support		
12.1	Is there a group of supporting donors for statistics?	
12.2	What are the main projects and programmes that will be funded by external support in the next few years and through which some activities of the DIL could be covered?	
12.3	What is the support that the NSO can expect to get from the national budget for its statistical programme in the next three years?	

			Total
			94
			100%

Annex 3: Presentation of the recommendations form the DIL diagnostic

Activities		Term
Supporting factors		
NSO's internal vision for the transformation of the data eco-system	Description of the intervention proposed to tackle the issue	To implement in the short, medium or long term
Human resources and training		
IT infrastructure		
Innovative experiences		
Quality		
Steering factors		
Legal and institutional framework		
Strategic framework		
Partnerships		
Communication		
Relations with the users		
Financing innovation		
Data analytics		

Annex 4: A format for the formulation of the NSO DIL strategic vision

Title: The transformative vision for the data ecosystem and for the installation and deployment of DIL activities

Executive Summary

1. Background to the proposal

- Brief description of the ecosystem and the major challenges facing its evolution.
- History of Government and NSO Efforts and Commitments on Data Innovation and Related Issues Results and main results of an initial diagnosis

2. Vision and objectives

2.1 Vision

A short statement in the form of a message that covers the ambitions in this area. Example:

«A statistical office at the center of national efforts to transform the data ecosystem in the country through multiple and effective partnerships promoting the development and dissemination of innovation as well as the valorization of data at all levels of the statistical value chain. »

2.2 Objectives

Articulation in the form of a diagram:

Global Goal (GG) (revolving around data ecosystem transformation) Specific Objectives (SO) 1, 2, 3 ... (ideally less than 5)

Each of the objectives must be the subject of a short explanatory text

2.3 Activities

Activities in each OS

Description

Activity 1.1,

Activity 1.2 ...

Activity 2.1,

Activity 2.2 ...

Example

Activity x.x: Motivate and train the staff of the Statistical Institute to make them more innovative The Statistical Institute must reflect on mechanisms for motivating staff that are linked to the development and implementation of innovation. The civil service statute does not leave much room for manoeuvre in terms of promotion and rewards, and it will therefore be necessary to innovate in this area. Above all, we must avoid the «brain-drain» to the private sector and the outside. We can envisage different levels of motivation of the staff which are related to the interest of the work related to innovation, the work environment, the possibility of increasing national and international visibility but also to participation in specific and rewarding initiatives. The distribution of time between business activities and other more rewarding activities (training in particular) could make it possible to establish job profiles and on this basis to develop personnel time management processes that open up room for innovation and training. The establishment of continuous training cycles for staff that are very targeted by groups and functions would be a plus, especially if it is part of an overall framework such as that of the creation of an integrated training centre. The IT staff should benefit from an update on data innovation tools and applications. The «innovation» approach that is looming beyond the DIL can be trying for the staff who may see it as a danger to their way of working and their professional profile. Raising awareness of data science should not be imposed but rather accompany the individual desire of each person at the Institute to integrate themselves into the transformation process that is being launched. The adjustment must be gradual, even if it is essential to initiate an effective transition adapted to the changes and challenges facing official statistics today.

2.4 Summary of objectives and activities

Specific objective 1 Title of the SO
1.1 Title of the activity 1.2 Title of the activity 1.3 Title of the activity ...
Specific objective 2 Title of the SO
2.1 Title of the activity 2.2 Title of the activity 2.3 Title of the activity ...
Specific objective ...
...

3. Installation of the DIL

3.1 Functions of the DIL Example

The functions to be assigned to the DIL are of several kinds:

- 1. Vision of an innovative/developer DIL. The objective is to experiment with innovation projects and then to make them a reality. These projects must contribute directly to the improvement of practices and the increase in the efficiency of the NSO.*
- 2. Vision of a catalyst DIL. The DIL is a tool for informing and centralizing the results of innovation experiments in the field of data science. It also participates in the dissemination of practices and the emergence of a data culture within the data ecosystem,*
- 3. Vision of a facilitating DIL. The DIL is a structure that connects innovation actors, builds and supports their networking. It is a force for proposing partnerships to boost innovation efforts and mobilize funding.*

These different roles each have their own usefulness, but also their own difficulties and methods of implementation; They must therefore be considered individually to properly structure the platform which will host and coordinate the various interventions. Nevertheless, there is a natural synergy between them as they reinforce each other; It is imperative to build bridges between them.

3.2 Establishment and implementation of the DIL in the short-term Example

The DIL should be established in a phased manner to accommodate the resource constraints (human and financial) faced by the NSO, on the one hand, and the degree of complexity of certain activities related to the transformation of the data ecosystem, on the other hand. While this transformation corresponds to a medium- and long-term vision, the implementation of a DIL is the founding element in the short term. It is a question of initiating a process of change that has broad and diverse ramifications. The establishment of a DIL is an essential dimension on the road to the transformation project and it must therefore be deployed quickly. Other factors will either support the efforts undertaken with the DIL (Supporting Investments) and others will guide and strengthen them (Steering investments). In concrete terms, this can be reflected in the diagram below:

(Diagram to be developed – reference to the section of the guide «What typical interventions can be deployed with a national DIL project? »)

There are therefore elements that can start quickly and that will feed into the work of the DIL. Others will only have effects in the medium or even long term. In the short term, and beyond the implementation of the DIL within the NSO, the following can be mentioned in particular:

Examples

This set of short-term activities is also presented in the annex. (see Appendix 3 of the guide)

3.3 Organisation

Example

The DIL is above all a set of functions and activities, but it is nevertheless necessary to give it a form that is visible and concrete within the NSO to accommodate the first developments of the innovation approach. It may not be necessary to create a new structure/unit in the first instance, but rather to designate one or more internal structures that coordinate the deployment of activities and/or the facilitation of a virtual platform.

4. Risks and mitigation measures

Risks identified	Mitigation measures
<i>Example: The human resources available at the Statistical Institute are limited. It may not be possible to mobilize the necessary expertise to launch the activities of the DIL.</i>	<i>The proposed approach for the creation of the DIL and the gradual ramp-up of its activities will make it possible to limit the initial commitment in terms of human resources. The training efforts will make it possible to strengthen the skills and knowledge within the NSO in terms of data science and therefore to multiply the capacities that can be mobilized. A major effort will be made to use the resources available in the regional directorates and to empower young people. Partnerships with universities and in particular the deployment of projects around Master and PHD works will contribute to the work while limiting the input of NSO staff.</i>
<i>Example: The responsibility of the Statistical Institute as the national lead agency for data collection within the NSS and beyond (non-conventional sources) is not effective in practice.</i>	<i>The proposal provides for several activities that are likely to raise awareness among partners of the need for interconnectivity of databases and secure exchanges within the NSS. Some aim to promote arguments to decision-makers but also to set up collaboration mechanisms with certain partners on an exemplary basis.</i>
<i>Example: The legitimacy of the Statistical Institute to take a leadership role in the formulation of a national data policy is not formally established.</i>	<i>In this area, the proposal provides to produce an advocacy document on the statistical institute's expertise in data innovation to be shared with its partners. It also provides for discussions within the NSS to raise awareness among its partners.</i>

5. Other themes of the proposal

(see section «What typical interventions can be deployed with a national DIL project?» and Annex 3 of the guide)

5.1 Partnerships Develop on the types of partnerships to be built: modalities, content, etc.

5.1.1 Research and Universities

5.1.2 Private Sector

5.1.3 Other partners

5.2 Other

To be developed according to priorities (examples: visibility and relationships with users, data analytics, etc.)

6. Monitoring and evaluation

6.1 Monitoring/evaluation of individual innovation «projects»

6.2 Overall monitoring/evaluation on the proposal

7. The positioning of the DIL in the support of statistics received by the country

(Description of major statistical projects with technical and financial support received under national and external funding)

7.1 Current Projects

7.2 Upcoming projects

8. Execution Plan

It is being developed over a period of 3 years, which should correspond to the time needed for the actual installation of the DIL. (see annex)

9. Financial projections

Financial estimates of the implementation of the project over 3 years are given in the funding table below. The total external financial requirement over the 3 years amounts to xxxxx. To be detailed (see appendix)

Execution Plan	Year 1				Year 2				Year 3				LT
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	
Specific Objective 1													
Activity 1.1													
Details of the activity													
Activity 1.2													
Activity 1.x													

Financing needs

(Estimates)		Year 1	Year 2	Year 3	Total
Specific objective 1					
Activity 1.1					
Total Specific objective 1					

Specific objective 2:					
Activity 2.1					
Total Specific Objective 2					

(Estimates)	Year 1	Year 2	Year 3	Total
Total SO 1				
Total SO 2				
Total SO x				
Total				

Annex 5: ICT reference framework

Topic	Sub-topic	Description of the sub-topic	Assessment question	minimum capacities required
Infrastructure	Resources	The essential (or even minimal) IT components in order to guarantee high availability of data (also services) and above all with a certain security are: - For intense calculations, you will need three servers - To protect access to servers, you need to have two firewalls (we opt for an open source and solid solution which is pfSense), one functional firewall and the other standby - To protect equipment (servers, switches...) against power cuts, you need to have an inverter - The servers give off heat, you will need two precision air conditioners, one functional and the other standby - For even more data protection, you need a backup solution (a server with low computing but a good storage capacity) - For the connections between all these equipments, you need to have a backbone switch (contains 40 BG or 100 GB ports or more). The second switch plays the role of load balancer. - A 10 GB switch for managing equipment from the outside. - As software, you just need to have a robust and open-source hypervisor that is proxmox If you need AI processing, you just need to put a good specification of the servers to acquire a GPU card or more like nvidia A100 or H100. These cards will be integrated into the servers through PCI express cards.		
	Strategy	Strategic planning to channel funding for the construction of a mini data center offering AI infrastructure	How can the funds obtained through different projects be channelled to build a mini data centre? What is the estimated budget for the data center? What are the long-term objectives?	Project management, strategic planning, knowledge of the IT sector, server specification

	Hardware specification	Specifications for servers, CPUs, storage technologies, networking equipment	What are the available compute resources (RAM and CPU)? Which storage technologies are right for our needs? How to support scalability?	Knowledge of hardware architectures, performance evaluation
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Topic	Sub-topic	Description of the sub-topic	Assessment question	Minimum capacities required
Infrastructure	Virtualization	Using virtualization to create virtual versions of computing resources, such as servers, storage devices, and networks. This allows for more flexible and efficient allocation of resources, better isolation and security, as well as simplified management and maintenance.	Which virtualization technologies are currently used within the infrastructure and for what needs? What hypervisor is used? Are there any network virtualization skills?	Knowledge of virtualization platforms: Proficiency in tools such as VMware, VirtualBox, Proxmox for creating and managing virtual machines. Experience with containers: Skills in using Docker, Kubernetes for container management, enabling efficient orchestration and scaling of applications. Network Virtualization: Understanding of network virtualization technologies. Configuration automation and management: Using tools like Ansible, Terraform to automate the deployment and management of virtual environments.
	Monitoring	Monitoring systems for the health, performance and security of the IT infrastructure.	What monitoring tools are used? How do you proactively manage incidents?	Skills in surveillance systems and data analysis
	Security	Cybersecurity measures to protect data and infrastructure.	What cybersecurity strategy for the data center? How to protect sensitive data?	Expertise in IT security

	Infrastructure AI	NVIDIA cards for AI computing, machine learning, and deep learning.	How is the AI infrastructure scaled? What is the evolution plan?	Knowledge of GPU computing, AI frameworks
	« Infrastructure as Code »	Automatic deployment and configuration of IT resources.	Which IaC tools will be used? Impact of IaC on efficiency?	Experience with Terraform, Ansible, other IaC tools
	Backup	Backup strategies and solutions to ensure the security, integrity, and availability of data in the infrastructure.	What is the current backup strategy and how is it implemented in the data infrastructure?	Knowledge of backup solutions: Experience with backup tools and platforms such as TrueNas

Topic	Sub-topic	Description of the sub-topic	Assessment question	minimum capacities required
Data science	Data Analysis	Inspection, cleaning, transformation, and data modeling techniques and processes to uncover useful insights, suggest conclusions, and support decision-making.	* Which tools are the most used data analytics solutions in your projects? * How do insights from data analytics influence decisions within the organization? * What are the main challenges faced during data analytics and how are they overcome? * Can you give a concrete example where data analytics has led to significant improvement or innovation?	*Proficiency in statistical techniques *Experience with data analysis tools such as Python (pandas, NumPy) or R * Ability to interpret data results and communicate insights effectively
	Machine Learning (ML)	Techniques and tools for machine learning.	What ML skills are available? Training needs?	Knowledge in statistics, Python, and ML libraries such as scikit-learn

	Deep Learning	Advance technics for deep learning.	What frameworks and tools for deep learning? Current or planned projects?	Experience with TensorFlow, PyTorch
	AI	Artificial intelligence applications and technologies.	How is AI integrated into current processes? Future projects?	Knowledge of AI principles, implementation of AI solutions
Development	Project management	Software Development Project Management.	What project management methodologies are used? Efficiency?	Project management skills, tools like JIRA
	Programming languages	Knowledge and proficiency in programming languages essential for the development of data science and software projects.	Which programming languages are currently mastered by the development team?	Python, R, Java, Scala

Topic	Sub-topic	Description of the sub-topic	Assessment question	minimum capacities required
Development	« Agile methodology »	Agile methods for project management.	Do you use agile methodologies for your project management?	Knowledge of agile methodologies, Scrum
	« DevOps »	DevOps practices for continuous integration and continuous delivery.	What DevOps tools and practices are in place? Planned improvements?	CI/CD skills, tools like Jenkins, GitLab
	« MLOps »	MLOps practices for deploying and managing ML models.	How are MLOps practices integrated? Impact on the life cycle of the models?	Knowledge of MLOps practices, experience with Kubernetes
	« Big Data Pipeline »	Big data collection, ingestion, storage, processing, and visualization.	How is the big data pipeline managed? Technologies used?	Data engineering skills, data pipeline tools
	Technologies Big Data	Explore key technologies used to process and analyze large volumes of data.	What big data technologies are currently used in your infrastructure, and for what types of applications or use cases?	Hadoop, Spark, Kafka, Zookeeper, Elastic search, Hive

Big Data	NoSQL	NoSQL databases for big data management.	What are the uses of NoSQL? Advantages over relational basics?	Knowledge of NoSQL databases, data modeling
	Big Data ML	Distributed processing to ensure the application of machine learning algorithms on distributed data	Do you know that there are specific technologies for ML on distributed data?	Apache Mahout, SparkML, FlinkML
Data Management	Data Catalog	Data Catalog for Research and Access.	Do you have a data catalog? How is the Data Catalog used? Accessibility and usefulness?	Experience with data cataloguing tools, metadata
	Data Quality	Maintaining data quality and reliability.	How is data quality ensured? Methods and tools?	Knowledge of data quality principles, analytical tools

Topic	Sub-topic	Description of the sub-topic	Assessment question	minimum capacities required
Data Management	Metadata management	Metadata management for context and lineage.	How is metadata management done? Importance for the organization?	Metadata management, using data catalogs
	Data Governance	Policies and procedures for data governance.	What are the data governance policies? Compliance and impact?	Knowledge of data governance, relevant regulations

Annex 6: Simplified model

The DIL maturity model is complex, and its implementation requires that there is already within the NSO a clear commitment to developing innovation. In an initial stage of reflection on innovation, a «quick assessment» can also be done. It builds on some key questions that may help assessing if favourable conditions exist to engage in the transformation of the data ecosystem. It may also help identifying what initial activities would be to be implemented as a priority to advance the transformative agenda. Questions could relate to the following aspects that cover the 12 dimensions of the full maturity model:

1. A leadership within the NSO convinced of the merits and on the necessity of the transformation and a «benevolent» adherence of management and staff to this vision.

Situation	Yes, it even exists partially	No, it does not exist or is very embryonic
Means of assessing the situation	- Review of statements by the DG/SG and NSO executives at internal management meetings, debates in the framework of the National Statistics Council (or equivalent coordination structure of the NSS) or at national and/regional/international events - Interviews with NSO management and some staff members	
Potential activities to be developed	- Consolidate what is being done internally for raising awareness but also through the search for strong support at the level of national decision-making (advocacy based on concrete results with the government) and at the level of regional and international donor and/or technical organizations (levers)	- Raise awareness of the NSO management and leadership on the issue of transformation through exchanges with other countries, participation in regional and/or international workshops

2. A strategic vision on the transformation to be undertaken is drafted at the level of the NSO and translated into an operational work plan that is implemented gradually according to the established schedule

Situation	The vision exists and the operational plan is being implemented	No vision, no operational plan
Means of assessing the situation	- Specific strategy paper on the vision for transformation or section in a broader strategy document (NSDS or other) - Short- and medium-term operational plan - Interviews with NSO management and some staff members	

Potential activities to be developed	- Regular evaluation of the progress of the implementation of the vision using performance and impact indicators. - Review and adjustment of the strategy in the light of acquired knowledge and experience.	- Engage in internal reflection on the transformative vision of the NSO through debates and discussions, to be gradually extended to partners in the national ecosystem. - Preparation of the strategic document and wide dissemination within the NSO. - Proficiency in statistics, Python, and machine learning libraries such as scikit-learn.
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3. A legal and institutional framework that allows for adjustments both to the structure of the NSO (internal organisation) and to its role as coordinator within the NSS and beyond for all questions relating to the use of «open data».

Situation	Permissive and flexible framework	Rigid frame and difficult to fit
Means of assessing the situation	- Statistics law - Legal framework governing the functioning of the National Statistical Council (or similar structure of NSS coordination) - Documents on the administrative and political organization of the country - Legal framework governing the structure and internal organisation of the NSO - Interviews with INS management and some staff members	
Potential activities to be developed	- At the internal level of the NSO, rethink the organization of work and the structure to best adapt it to the requirements of the transformation. - Consolidate the NSO in its role as coordinator of the NSS and expand to the entire ecosystem through a specific decree regarding the use and sharing of non-statistical data.	- To make a comprehensive diagnosis of the legal and institutional framework governing statistical coordination and coordination for the use and sharing of non-statistical data. - Describe in detail the internal organization of the NSO and look for points for the convergence of activities related to data science.

4. Staff trained and/or aware of data science within the NSO

Situation	There are staff with this type of skill	There are no staff with this type of skill
Means of assessing the situation	- Documents on human resources within the NSO (quantity and quality) - NSO Training Programs - List of national (and regional) organizations teaching data science and review of their curricula - Interviews with the Human Resources and Training departments at the NSO	

Potential activities to be developed	- Use staff trained in data science skills to multiply knowledge within the NSO but also beyond among partners. - Establish long-term partnerships with schools and/or universities that provide training in data science (practical teaching by NSO staff, hosting interns and diploma work within the NSO. - Ensure regular updates through refresher training and/or participation in regional and international meetings and workshops on the subject.	- Mapping of existing skills internally - Data Science Needs Profile - Training and/or recruitment plan for skills in the field
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5. Modern and powerful IT equipment within the NSO

Situation	Recently acquired/modernized powerful modern hardware and up-to-date licenses on recent analysis and processing applications and software	Low-powered hardware that does not allow the use of modern and sophisticated software
Means of assessing the situation	- Review of existing IT hardware (equipment and software) at NSO - Interviews with IT and Development Managers	
Potential activities to be developed	- Mapping of work to modernize data collection, processing, analysis and dissemination techniques within the NSO - Up-grading of practices on certain specific aspects (security, data transfers, etc.)	- Mapping of IT and application modernization needs and modernization plan - Mapping of the training needs of NSO IT executives and training program

6. Achievements/experiences in the use of administrative data for statistics

Situation	There are several experiments underway with different partners	Nothing concrete has been done in this area
Means of assessing the situation	- Internal NSO documents (partnership agreements, MoU, etc.) - Review of the methods of processing and use of registers and administrative data at the NSO - Interviews with the INS staff involved and some partners	

Potential activities to be developed	- To draw lessons from current experiences and identify good practices - Programme to support partners to disseminate good practices and raise awareness of quality - Consolidation of the most promising partnerships and gradual commitment to collaboration with new partners - Documentation of experiences and dissemination/sharing at the national as well as regional and international levels.	- Dissemination of experiences from other countries and context in this field and discussion of their potential replication - Identification of the most promising partners and progressive commitment to collaboration with them for the processing of administrative data
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7. Achievements/experiences in the use of other non-statistical sources

Situation	There are several experiments underway with different partners	Nothing concrete has been done in this area
Means of assessing the situation	- State of play of non-statistical data sources in the country - Internal INS documents (partnership agreements, MoU, etc.) - Interviews with NSO staff involved and some partners	
Potential activities to be developed	- To draw lessons from current experiences and identify good practices - Consolidation of the most promising partnerships and gradual commitment to collaboration with new partners - Documentation of experiences and dissemination/sharing (at the national level as well as at the regional and international levels)	- Dissemination of experiences from other countries and context in this field and discussion of their potential replication - Identification of the most promising sources and progressive engagement in the mobilization and use of these sources

8. Organized and articulated partnerships with NSS actors and owners of non-statistical data sources in the private sector

Situation	Partnerships exist and are effectively managed by the NSO	Partnerships are developed on an ad hoc basis without overall coherence
Means of assessing the situation	- Internal NSO documents (partnership agreements, MoU, etc.) - Interview with the administrative services of the NSO	

Potential activities to be developed	- Consolidation and updating of partnerships incorporating precise protocols for collaboration and exchange - Secure data transfers - NSO Advice and Assistance Program for the Benefit of its Partners	- Articulation of ongoing partnership experiences within an overall strategic framework - Mapping of partners' needs in terms of technical support and statistical training
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9. Well-defined internal considerations for the application of regional and international norms and standards as well as for quality

Situation	A coherent strategy exists within the NSO for quality monitoring and the implementation of norms and standards	There are practices in progress but no overall plan on these issues
Means of assessing the situation	- Repository of methodological information (GDDS, SDDS) - Internal NSO documents on quality and application of standards - Interviews with NSO staff members involved in these areas	
Potential activities to be developed	- Review of online achievements and lessons to be learned for the future - Extension of the review work to the situation in NSS partners - Consolidation plan	- Raising awareness of the benefits of taking quality into account at all levels of the NSO's work - Documentation (guides) on the application of the standards at the NSO

10. Demonstrated concern for effective consideration of user needs

Situation	Users are regularly consulted on the satisfaction of their needs through different instruments	There is no regular and systematic contact with users
Means of assessing the situation	- Social networks and website of the NSO - Internal documents at the NSO (dissemination, communication, etc.) - Interviews with the staff responsible for dissemination and communication at the NSO	
Potential activities to be developed	- Consistency of the various tools for collecting user perceptions and satisfaction levels (surveys, direct queries) - Mapping of uses and review of achievements - Consolidation of communication and dissemination practices and tools for the relationship with users - Specific media discussion programme	- User Mapping - Reflection on dissemination and communication with a view to the production of a strategy document and an associated operation plan

11. Proven and recognized expertise in research and complex analysis at the INS level and/ or in relation with research centers or academics.

Situation	The NSO is very open to scientific cooperation	The NSO focuses on the production and dissemination of data through their analysis
Means of assessing the situation	- Internal research documents at the INS, analyses and complex studies carried out - Interview with INS staff involved in research and with partners in local research centers and at the university	
Potential activities to be developed	- Consolidation of experiences and evaluation of results	- Study on the benefits to be gained, the modalities, and the constraints related to an engagement in analysis. - Action plan

12. A reservoir of active technical and financial support to consider interventions in support of the transformation of the data ecosystem

Situation	The NSO is supported by a group of loyal donors who are convinced of its competences with whom it maintains a lasting and transparent relationship	The NSO receives funding on an ad-hoc basis from various donors depending on the type of operation to be financed
Means of assessing the situation	- Internal NSO documents concerning the external support received in recent years and relations with donors and international technical bodies involved in statistical support - Interview with the INS staff involved in this field. Contact with the Ministry of Foreign Affairs and donors potentially interested in supporting the INS for innovation	
Potential activities to be developed	- Present to the group reports on the impacts of the contributions received - Argue for support to be given to the «core programme» of the INS and not to specific operations	- Establish a group of donors for statistics to be convened regularly and fed with strategic elements justifying operations and related funding requests

Annex 7: Comprehensive overview of AfDB Data Innovation Lab activities (2022–2025)

Comprehensive overview of AfDB Data Innovation Lab activities (2022–2025): Alignment with objectives, pillars, and guiding principles

Main Activities	Linked DIL Objective(s)	Implementation Pillar(s)	Guiding Principle(s)	Description & Achievements
“Distance” Accessibility Model (Meta)	Obj. 1 – Enable innovation Obj. 2 – Produce augmented & cost-effective data	Pillar 1 – Innovation for value & impact	Data Augmentation, Collect Once, Use & Reuse, Reduce Friction	Combined Meta HRSL and routing algorithms to identify underserved areas for health & education; informed national infrastructure planning.
Broadband Inclusion Mapping (Ookla + Meta)	Obj. 1 & 2	Pillar 1	Data Augmentation, Reduce Friction, Partnerships	Merged Ookla speed-test data with Meta settlement grids to reveal under-connected clusters and guide digital-inclusion policies.
Nighttime Lights Analysis – Burkina Faso (NASA)	Obj. 1 & 2	Pillar 1	Data Augmentation, Collect Once, Use & Reuse	Used VIIRS satellite radiance (351 communes) to map energy and infrastructure gaps; results integrated into the Country Strategy Paper.
Education Accessibility Mission – Rwanda (NISR)	Obj. 1 & 2	Pillar 1	Data Augmentation, Partnerships	Applied distance-matrix and geospatial analysis to evaluate school access in Nyagatare District; produced evidence for inclusive education planning.
Kigali Workshops 2023 – Big Data Technologies & Healthcare Accessibility	Obj. 3 – Demystify Big Data & Data Science Obj. 2 – Augmented data	Pillar 1 & 2	Data Augmentation, Reduce Friction, Partnerships	Two flagship regional workshops co-hosted with COMESA, NISR & PARIS21: • <i>Data in Action: Big Data Technologies and Open Semantic Resources as Enablers</i> (15–19 May 2023, 15 countries). • <i>Data-Driven Healthcare: Innovating with Big Data Technologies</i> (28 Aug–1 Sep 2023, 16 countries). Showcased an open-source architecture for healthcare accessibility in Kigali demonstrating efficient, data-driven health planning.

Capacity-Building Workshops (2024–2025)	Obj. 3	Pillar 1 & 2	Reduce Friction, Partnerships	Hands-on regional trainings with IMF and partners on Big Data infrastructure (Hadoop, Spark, Kafka, ELK), geospatial analytics, AI workflows, and LLM/prompt-engineering.
DIHAP 2024 – Data in Health (AfDB – World Bank)	Obj. 3	Pillar 1 & 2	Data Augmentation Partnerships	Blended learning program (virtual + in-person in Tunisia) training 48 participants from 13 countries on Big Data for health.
Integrated Health Data Framework – “Connected Health, Informed Decisions”	Obj. 1 – Innovation Obj. 2 – Augmented data Obj. 3 – Demystify Big Data Obj. 4 – Open tools	Pillar 1 & 2	Data Augmentation, Collect Once Use & Reuse, Reduce Friction, Partnerships	Developed a multilingual, facility-centric health-data model and open-source architecture (web app + dashboards + map layers). Delivered insights on facilities, staff, stock-outs and disease trends; a replicable blueprint for RMC health analytics.
Brown Bag Webinar Series (Sessions 1 & 2)	Obj. 3 & 4	Pillar 2	Partnerships	<i>S1 (21 Mar 2024): Essence of Data Partnerships (Meta, WB); S2 (30 Jul 2024): Data in Action (NISR, LinkedIn, universities).</i> 117 participants shared practical use cases.
Innovate Africa Symposium (2023 Johannesburg / 2024 Kigali)	Obj. 3 & 4	Pillar 2	Partnerships, Reduce Friction	Continental flagship forum on data governance and innovation; 2024 edition (127 participants, 22 countries) produced a Pan-African Data Governance Framework and network of practitioners.
Side Event – 54th UN Statistical Commission (UNHQ, Mar 2023)	Obj. 1 & 3 & 4	Pillar 2	Partnerships	“Network of Data Innovation Labs in Africa” event gathered African NSO leaders and partners to present the DIL vision and build continental collaboration.
Francophone & Anglophone DIL Networks	Obj. 1 & 3 & 4	Pillar 2	Partnerships, Collect Once Use & Reuse	Regular bilingual cluster meetings (country focal points + working-group leads + AfDB DIL team). Agenda: country updates, deep dives on use cases, governance/quality checkpoints (documentation, reproducibility, ethics), and forward look on training

				and partnership needs.
Participation & Secretariat Role – STG17 on Emerging Issues (AU STATAFRIC)	Obj. 1 & 3 & 4	Pillar 2	Partnerships	AfDB served as technical facilitator and Secretariat of STG17 (Kigali, 10–12 Sep 2025). Brought together 13 countries and RECs to advance innovation in statistics, share Data Lab experiences, and adopt an Action Plan (2026–2030) for leveraging Big Data and AI in official statistics.
Data Ecosystem Diagnostics (7 countries)	Obj. 1 & 2	Pillar 1 & 2	Collect Once Use & Reuse, Reduce Friction, Partnerships	Applied 12-dimension maturity model in Tunisia, Senegal, Côte d’Ivoire, Nigeria, Cameroon, Zambia, Botswana to inform Country Action Plans.
MDLAs signed (Meta & JBA)	Obj. 1 & 4	Pillar 2	Partnerships, Data Augmentation	Formal access to Meta Data for Good and JBA flood-risk datasets for analytics on accessibility and climate resilience.
Negotiations in progress (Google & Ookla)	Obj. 1 & 4	Pillar 2	Partnerships	Advanced negotiations to expand access to broadband, mobility, and network-performance datasets.
Participation in DDP Day 2025 (Google HQ)	Obj. 1 & 4	Pillar 2	Partnerships	Represented AfDB at global DDP forum in New York; highlighted Africa’s leadership in responsible data use and collaboration.
Open Repository of Tools & Methods	Obj. 4	Pillar 1 & 2	Collect Once Use & Reuse, Reduce Friction	Curated open-source pipelines, dashboards, and visualization modules for reuse by AfDB and RMC partners.